



Affirmed Private Network Service Manager Deployment Guide

Release 1.3.0

Date: February 2023

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Preface

This document describes how to

- Deploy Private Network Service Manager and ASAP in Azure Cloud
- Deploy Private Network Service Manager and ASAP VM - HA Mode
- Deploy Private Network Service Manager and ASAP VM - non-HA Mode
- Deploy Netfoundry Zero-Trust Network
- Upgrade Service Manager
- Deploy APNS in Cloud

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for installing, provisioning, and monitoring the system. The procedures in this guide require you to understand and follow the safety practices at your site, as well as those identified in this guide.

How to Use This Guide

This guide contains the following chapters and appendices:

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menu	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .

Description	Convention	Examples
Selecting a menu item	Menu => Option	Select Security => Group Management
Variable parameters for which you supply values	<courier bold italic>	<i>configure_dr --host <host name></i>
User input	Courier bold font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular font	result success
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Affirmed Private Network Service Manager Deployment Guide</i> . <i>Workflow</i> is an application running on the network element.
Images in online help		Many images can be enlarged by clicking the image with the left-mouse button. Alternately, they can be enlarged/reduced by pressing the Enter key when the image is highlighted (surrounded by a thick border) on the screen.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

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Getting Started

Before we begin, it is recommended to go through the following terms used in this guide.

Azure Cloud

Azure is a public cloud computing platform with solutions including Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) that can be used for services such as analytics, virtual computing, storage, networking, and much more.

Azure Resource

A manageable item that is available through Azure. Virtual machines, storage accounts, web apps, databases, and virtual networks are examples of resources.

Azure Resource Group

A container that holds related resources for an Azure solution. The resource group includes those resources that you want to manage as a group.

Azure Resource Manager Template

The Azure Resource Manager (ARM) template is a JavaScript Object Notation (JSON) file that defines the infrastructure and configuration for your project. The template uses declarative syntax, which lets you state what you intend to deploy without having to write the sequence of programming commands to create it. In the template, you specify the resources to deploy and the properties for those resources.

Azure Stack Edge

Azure Stack Edge is a cloud-managed, hardware-as-a-service solution shipped by Microsoft. It brings the Azure cloud's power to a local and robust server that can be deployed virtually anywhere local AI and advanced computing tasks need to be performed.

Endpoints

An endpoint is node on the edge of your Network. Protected traffic flows to, from, and through Endpoints. Endpoints may be configured to dial or host Services, or both.

AppWAN

An AppWAN is a policy that controls access to Services and it works like a permission group. In the web console, An AppWAN is visually represented as Endpoints on the left that are allowed to connect to Services on the right.

Edge Router

Edge routers are specialized routers that act as gateways at the network edge. They enable connectivity between an enterprise network and an external network.

Netfoundry VM

The NetFoundry VM is a deployable system image with pre-installed software for a variety of virtual network functions e.g. customer-hosted Edge Router, gateway tunneler, Linux router. NetFoundry offers a variety of virtual machine system image formats that can be imported in your preferred VM stack or launched by on your preferred cloud.

The best way to obtain the latest VM for your Network is to create an Edge Router and then visit its detail page in the console and click “Get the VM” and follow the stack-specific instructions found there.

Zero-Trust Network

Netfoundry VM is built on the foundation of zero-trust network. The main concept behind zero-trust is that networked devices, such as laptops, should not be trusted by default, even if they are connected to a managed corporate network such as the corporate LAN and even if they were previously verified. Instead, the zero-trust networking approach advocates mutual authentication, including checking the identity and integrity of devices irrespective of location, and providing access to applications and services based on the confidence of device identity and device health in combination with user authentication.

High Availability Mode

High Availability (HA) Mode is the mode in which system can operate continuously without failing for a designated period. HA works to ensure a system meets an agreed-upon operational performance level.

Affirmed Private Network Service Manager (APNSM) Deployment Network Architecture

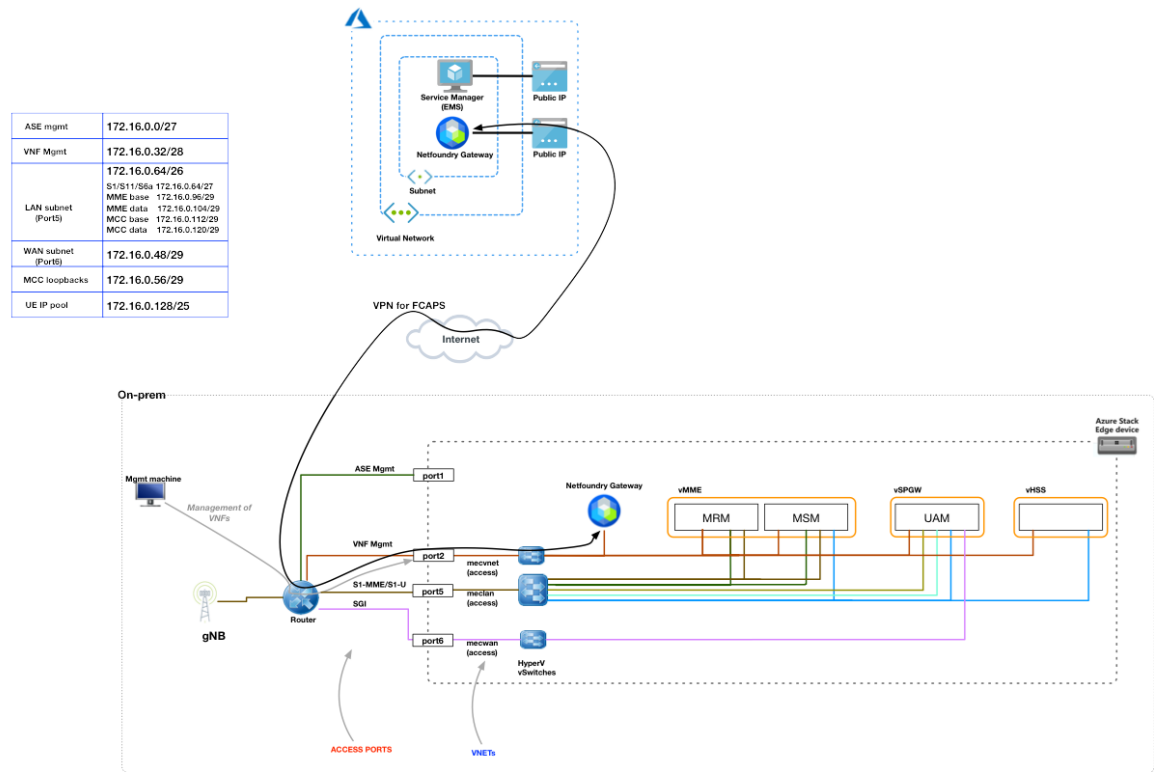
Azure provides Azure Stack Edge to bring Azure cloud's power to our private network.

Private Network Service Manager resides on Azure public cloud whose purpose is to deploy, configure, and monitor the packet core network and the network elements or VNF in the Azure Stack Edge.

To ensure that communication between Private Network Service Manager and Network Functions is fast, safe, and encrypted, it is necessary to deploy Netfoundry VM as the medium between both of them. The Netfoundry VM has two gateways: one on the Azure public cloud and the other one on the Azure Stack Edge.

The following diagram depicts the network architecture for Affirmed Private Network Service Manager (APNSM) deployment.

Figure 2-1 APNSM Deployment Network Architecture



Deploying Private Network Service Manager and ASAP in Azure Cloud

Note: The ARM templates (.json files) are included in the release package.

This chapter describes how to deploy Private Network Service Manager and Affirmed Service Automation Platform (ASAP) in Azure Cloud using ARM Template.

Prerequisites

- Azure CLI must be installed on your system and ensure that the CLI session is authenticated. To authenticate the session:
 - i. Run Windows PowerShell in Administrator mode.
 - ii. Execute the “**az login**” command. You will be directed to the Azure user authentication webpage.
 - iii. Enter your credentials. Your CLI session will be authenticated.
- If Azure CLI is not installed on your system, refer to [Azure CLI Installation](#).
- You must be the owner of the resource group you are deploying.
- A pre-existing Management Network is required.
 - **Virtual Network:** A Bastion subnet is recommended, but not required. The network space must be large enough to contain all devices you wish to connect to the management network. A /24 or larger is recommended. The virtual network does not need to be in the same resource group as the resources you wish to provision, but it must be in the same subscription.
 - **Virtual Subnet:** The network space must be large enough to contain all devices you wish to connect to the management network, a /27 or larger is recommended.
 - **Network Security Group:** It is recommended to simplify port connectivity. Minimum requirement that the management network must allow: tcp/443 and tcp/8443

Deploying the Infrastructure ARM template

The Infrastructure ARM Template deploys a virtual network and management subnet that can be used for the APNSM deployment, if user does not already have a suitable network setup. A network security group and Storage Account are also created for convenience.

Note: This is an optional procedure.

To deploy infrastructure ARM template:

1. Refer to the following infrastructure parameters file: *infrastructure-parameters-1.1.0.0-1*.
2. In the sample infrastructure parameters file given above, update the following parameters as per the requirement.

Table 3-1 Infrastructure Template Parameters

Parameter Name	Description
deploymentName	String prepended to all resource names to help in tracking a specific deployment. For example, apnsm. Must be lower-case letters and numbers only.
vnetPrefix	Subnet/Netmask to assign to the required management network. For example, 10.12.0.0/16
mgmtSubnetPrefix	Subnet/Netmask assigned to the required management subnet must be within network subnet. For example, 10.12.1.0/24.
publicIpAllowed1	IP Address that is granted access to the ServiceManager/SA application when connected to VPN. For example, whatsmyip.com when connected to VPN.
publicIpAllowed2	Second IP Address that is granted access to the ServiceManager/SA application when not connected to VPN

3. Refer to the following infrastructure template file: *infrastructure-1.1.0.0-1.json*
4. Using Azure CLI, execute the below command.


```
nohup az deployment group create --resource-group PLTE_Service_Manager --template-file infrastructure-1.1.0.0-1.json --parameters infrastructure-parameters-1.1.0.0-1.json > output.log &
```

where, PLTE_Service_Manager is the existing resource group name.

Private Network Service Manager and ASAP VM Deployment-HA Mode

Overview

Deployments requiring a higher degree of uptime and reliability should deploy the HA solution. In this deployment, both PNSM and ASAP instances are deployed as dual-node HA clusters,

where one node acts as the active server and the second node acts as the standby server. Standby nodes are automatically kept up to date in real time by MySQL database replication. In the event of a failure on the active node, the secondary node will take over the active role automatically.

In order to facilitate seamless communications regardless of whichever node is active, each cluster maintains a single external public IP address that is assigned to the current active node, and in the event of an active node failure is automatically moved to the new active node. This public IP address is statically assigned and will not change over the lifetime of the deployment.

Common Characteristics of Service Manager HA Deployment:

- Deploying the PNSM Application deploys the PNSM HA Cluster and an ASAP HA Cluster.
- Each HA cluster has 2 VMs.
- All 4 VMs use Static IP assignments on the same internal/private management network.
- Both the ASAP HA Cluster and the PNSM HA Cluster use a statically assigned Public IP.
- Geo Redundancy is not supported.

ASAP Characteristics:

- HA Cluster with 2 VMs, one Active and one Standby.
- MySQL Database replicated between VMs.
- One Private IP per VM for intra-cluster communications.
- Pacemaker controlled Virtual IP for internal communication with PNSM.
- One Public IP Address is created for the cluster (Static IP Assignment).
- Both the Private IP and the Public IP are automatically moved to the Active VM.

PNSM Cluster:

- HA Cluster with 2 VMs, one Active and one Standby.
- MySQL Database replicated between VMs.
- One Private IP per VM for intra-cluster communications, and communication with ASAP.
- No Pacemaker and no private Virtual IP.
- One Public IP Address is created for the cluster (Static IP Assignment).
- The public IP address is automatically moved to the Active server.

Note: Private Network Service Manager High Availability deployment mode is highly recommended for production.

Prerequisites

Private Network Service Manager and ASAP vTransactor deployment rely on multiple artifacts that must be in place before deploying the VM.

Before Private Network Service Manager deployment:

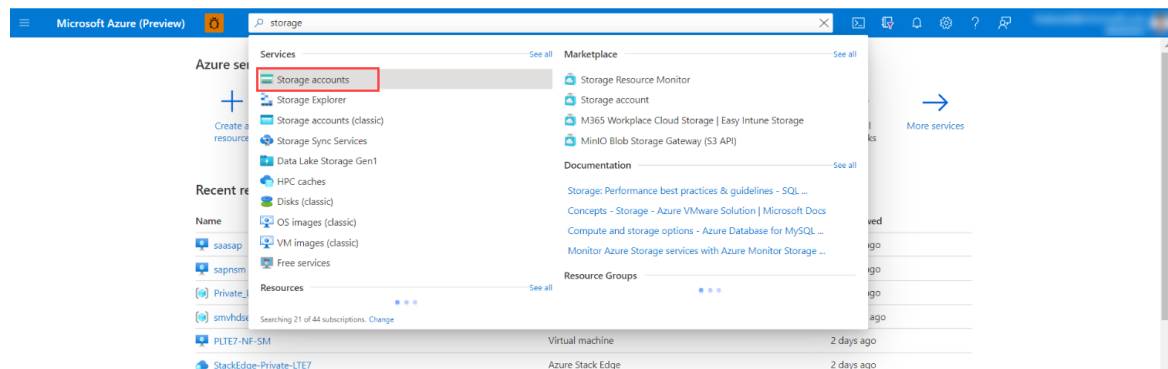
- Ensure that below mentioned files are uploaded to Azure Storage.
 - PNSM artifacts such as rpm file, ARM templates, EMS patch file, and EMS MySQL vhd file.
 - VNF vhd files for SAEGW, MME, and HSS.
 - Upgrade and Rollback image files for SAEGW, MME, and HSS.
- Ensure that the following Azure resources are created during the infrastructure deployment.
 - Storage account with specific containers, such as images, patches, rpms, and scripts
 - Virtual Network
 - Management Subnet
 - Network Security Group
- Generate SAS URL

***Note:** SAS URL is a read-only URL for a specific file that has a built-in expiration date. A SAS URL has following format:
<URL><SAS_TOKEN>
The image File location is NOT a SAS URL.*

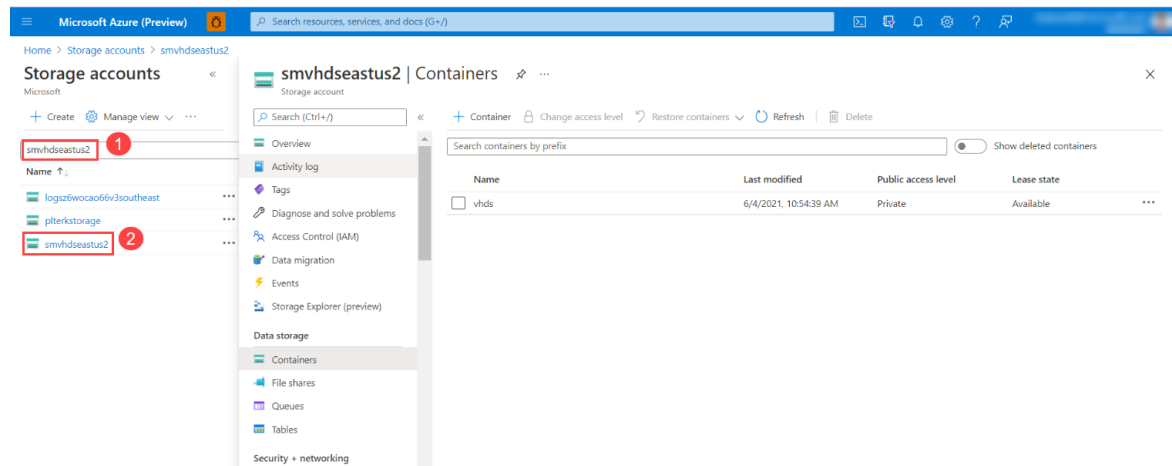
To generate SAS URL:

1. Go to search bar, type "storage accounts", and click **Storage accounts** from the dropdown. The Storage Accounts page opens.

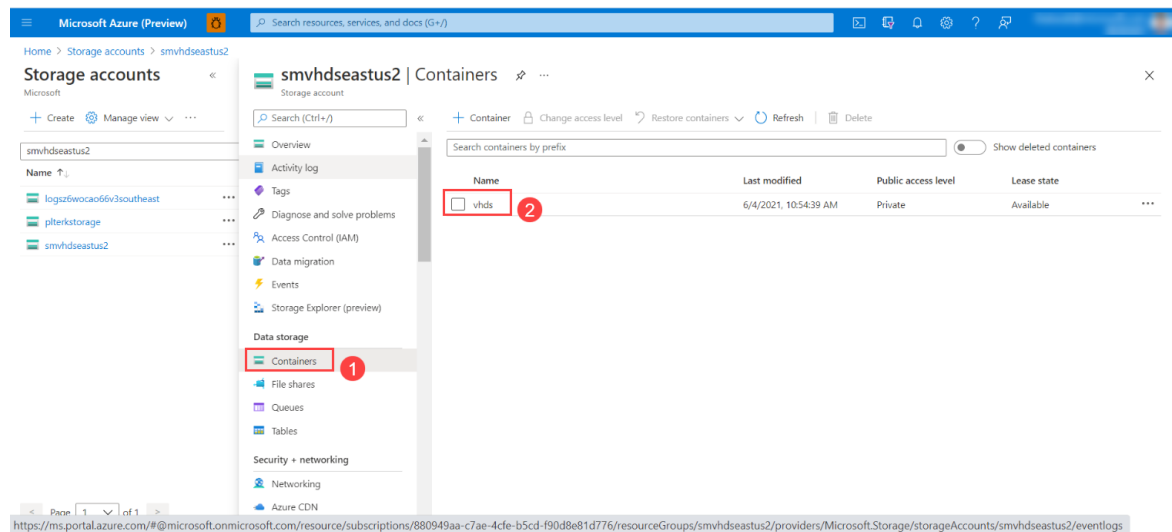
Figure 3-1 Opening Storage Accounts



2. Type account name in the search bar and click the respective storage account.

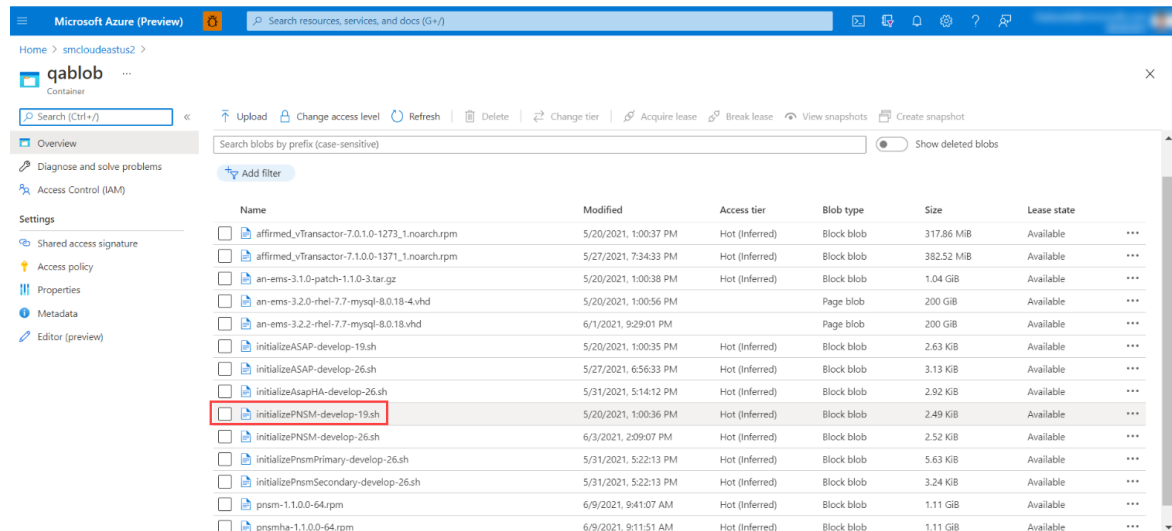
Figure 3-2 Clicking Storage Account

3. Under the **Data storage** section, click **Containers**, and then click the container.

Figure 3-3 Clicking Containers

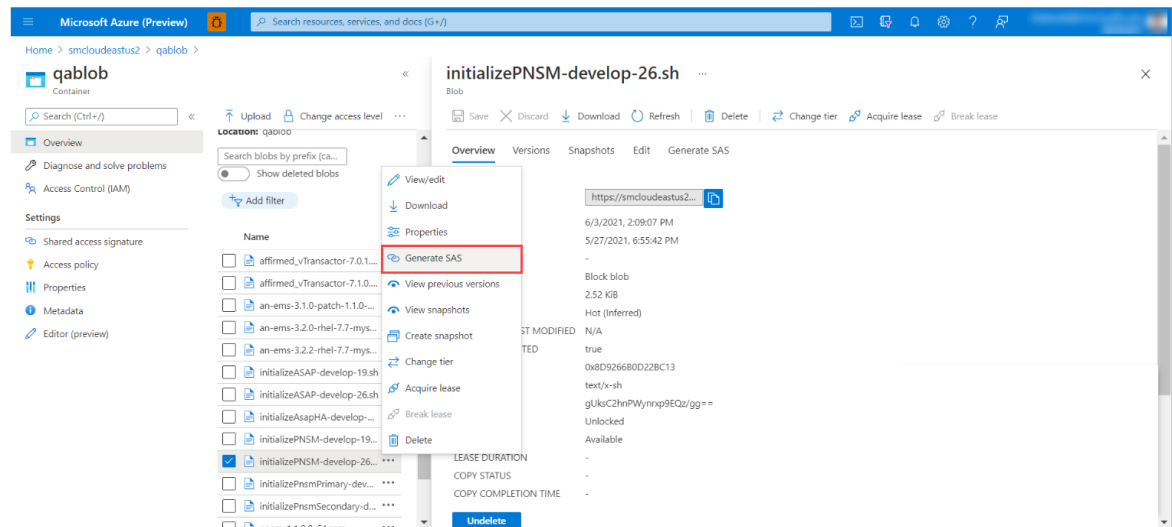
4. Click the required container and click the respective file.

Figure 3-4 Clicking Container File

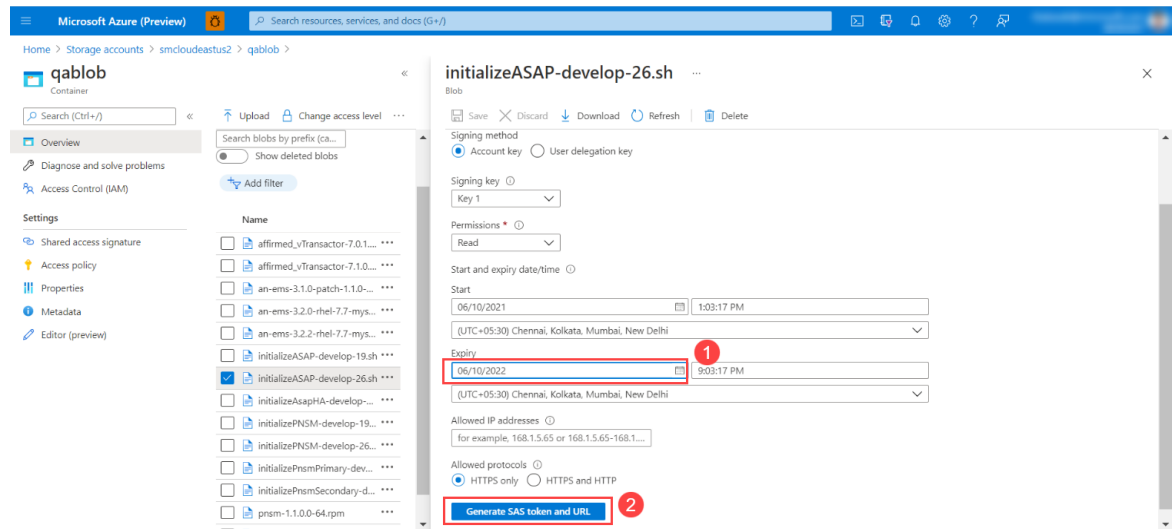


5. Right-click the file, and select **Generate SAS**.

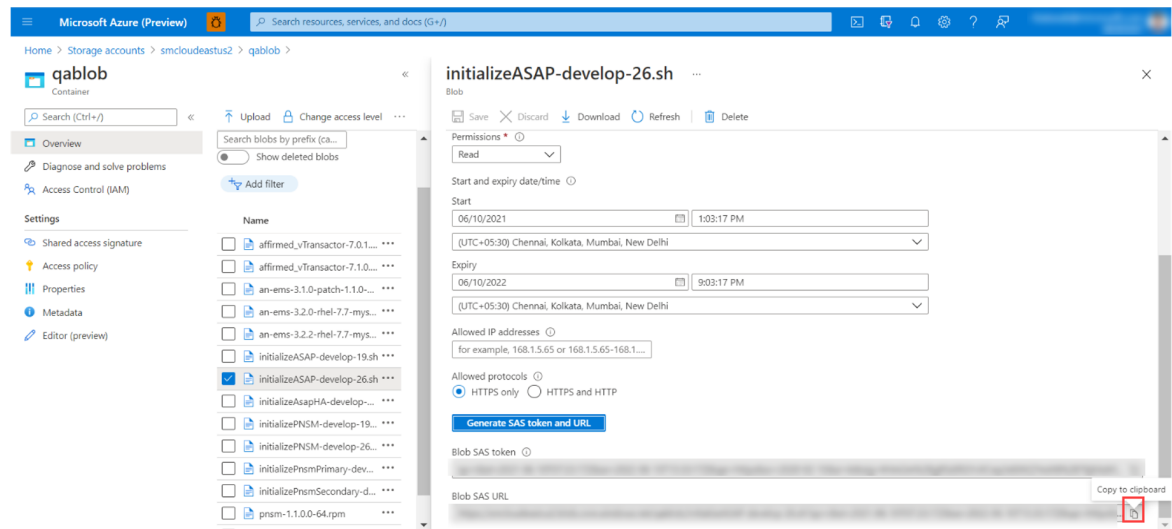
Figure 3-5 Generating SAS



6. Set the **Expiry** field and click **Generate SAS token and URL**. The SAS token and URL is generated.

Figure 3-6 Setting Expiry and Generating SAS Token and URL

7. Copy Blob SAS URL and use it as per the requirement.

Figure 3-7 Copying Blob SAS URL

Deploying Private Network Service VM

To deploy Private Network Service Manager VM:

1. Refer to the following Private Network Service Manager parameter template file: *pnsim-ha-parameters-1.1.0.0-1.json*.
2. In the sample Private Network Service Manager parameter template file given above, update the following parameters as per the requirement.

Table 3-2 Private Network Service Manager Template Parameters for HA-Mode

Parameter Name	Description
asapName	Root namespace used to create ASAP resources.
pnsmlName	Root namespace used to create PNSM resources.
asapVmSize	ASAP VM Size. See template for supported versions.
pnsmlVmSize	PNSM VM Size. See template for supported versions.
asap1PrivateIP	Private IP static assignment from the Management Subnet.
asap2PrivateIP	Private IP static assignment from the Management Subnet.
asapPrivateVIP	Private Virtual IP static assignment from the Management Subnet.
pnsml1PrivateIP	Private IP static assignment from the Management Subnet.
pnsml2PrivateIP	Private IP static assignment from the Management Subnet.
asapVmImageUrl	URL for the base VM image.
pnsmlVmImageUrl	URL for the base VM Image.
asapRpmSasUrl	SAS URL for the ASAP Application RPM.
pnsmlRpmSasUrl	SAS URL for the PNSM Application RPM.
asapOsPatchSasUrl	SAS URL for the ASAP OS Patch.
pnsmlOsPatchSasUrl	SAS URL for the PNSM OS Patch.
asapInitScript1SasUrl	SAS URL for the first ASAP init script. (e.g. initializeASAP.sh)
asapInitScript2SasUrl	SAS URL for the second ASAP init script. (e.g. initializeAsapHa.sh)
pnsmlInitScript1SasUrl	SAS URL for the first PNSM init script instantiated. (e.g. initializePnsmlPrimary.sh)
pnsmlInitScript2SasUrl	SAS URL for the second PNSM init script. (e.g. initializePnsmlSecondary.sh)
managementNetwork	Name of the Management Network to join.
managementSubnet	Name of the Management Subnet to join.
prefix	The netmask for the management subnet, e.g. 24, 27, etc
adminUsername	Username for the Linux non-root administrative user to be created. <i>Note: Don't provide the username such as, "admin", "affirmed", etc.</i>
adminPassword	Provide the admin password.
mysqlPassword	Provide the MySQL password.

Parameter Name	Description
repPassword	MySQL replication or asap password.
vaultPassword	Secrets vault password (ASAP).
grafanaPassword	MySQL grafana password (ASAP).
certPassword	SSL certificate password (ASAP).
keystorePassword	SSL keystore password (ASAP).
sbKeystorePassword	SSL southbound keystore password (ASAP).
haclusterPassword	hacluster linux user password. (ASAP)

3. Refer the following Private Network Service Manager template file: *pnsn-ha-1.1.0.0-1.json*.

4. Using Azure CLI, execute the below command.

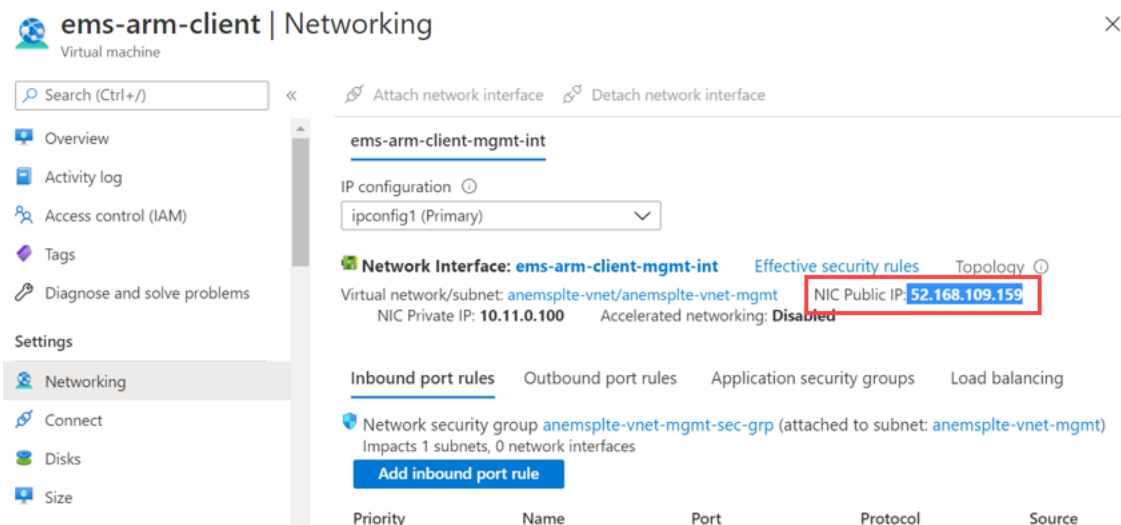
```
nohup az deployment group create --resource-group PLTE_
Service_Manager --template-file pnsn-ha-1.1.0.0-1.json --
parameters pnsn-ha-parameters-1.1.0.0-1.json > output.log &
```

Verifying Private Network Service Manager VM Deployment

To verify the Private Network Service Manager VM deployment, log in to Private Network Service Manager. To log in:

1. Go to the Azure portal and copy the **NIC Public IP Address** from the Private Network Service Manager deployed in the previous section.

Figure 3-8 Copying the Public IP Address



2. Open a new tab in your browser, type "https://", paste the IP Address in the address bar, and then press **Enter**. The Private Network Service Manager login page appears.

Configuring Private Network Service Manager VM

To configure Private Network Service Manager:

1. Provide the following credentials.

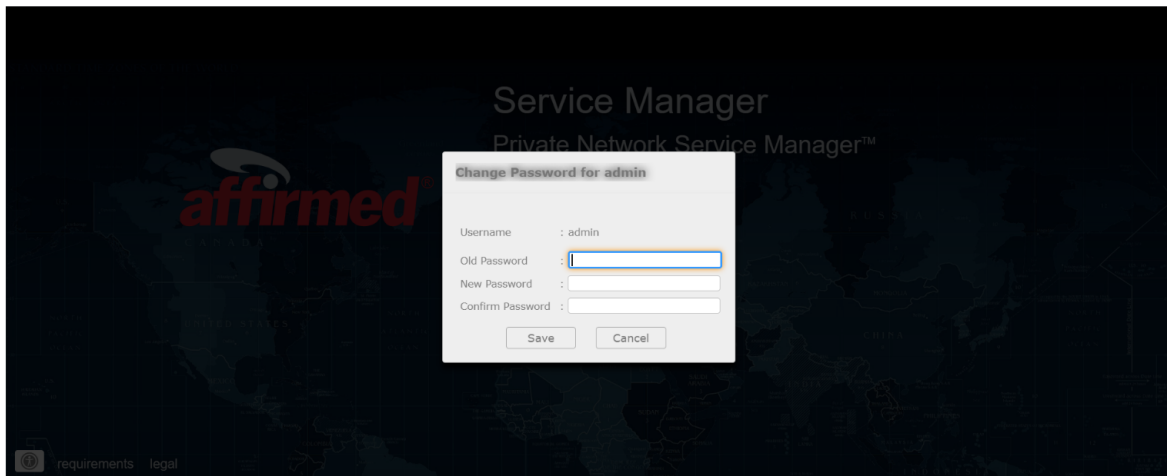
Username: xxxxxx

Password: xxxxxx

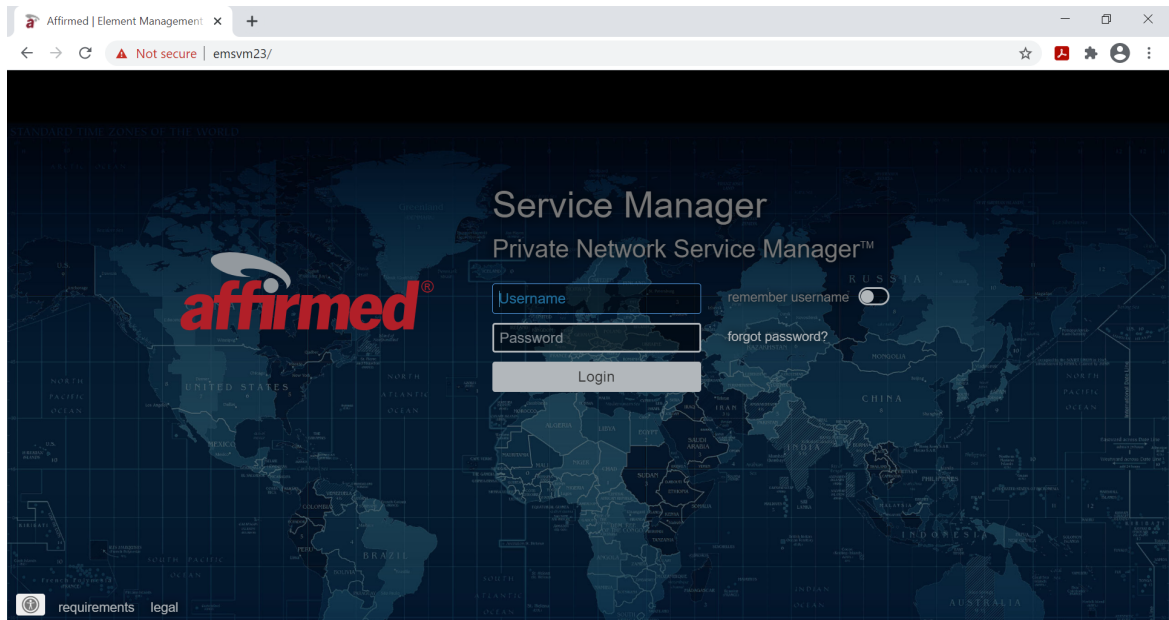
***Note:** For security reasons, username and password cannot be provided in the document.*

2. A page appears prompting you to change the password.
 - i. Provide **Old Password**.
 - ii. Provide **New Password**.
 - iii. Provide **Confirm Password**.
 - iv. Click **Save**.

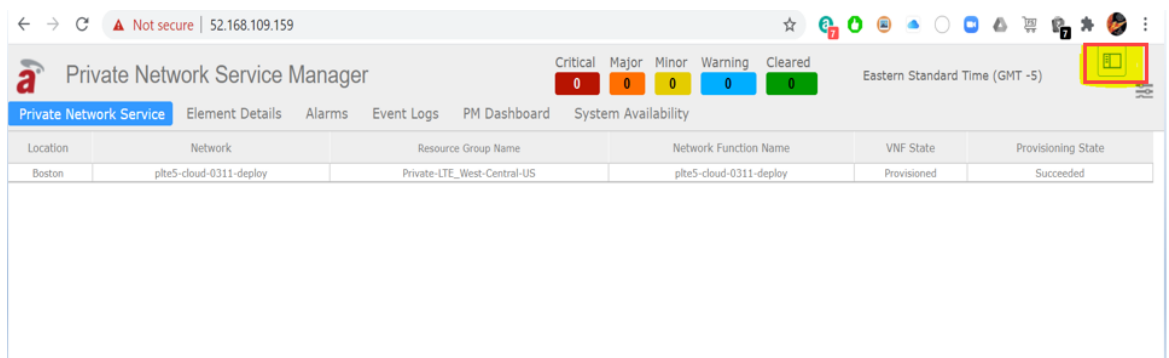
Figure 3-9 Changing Password for Login



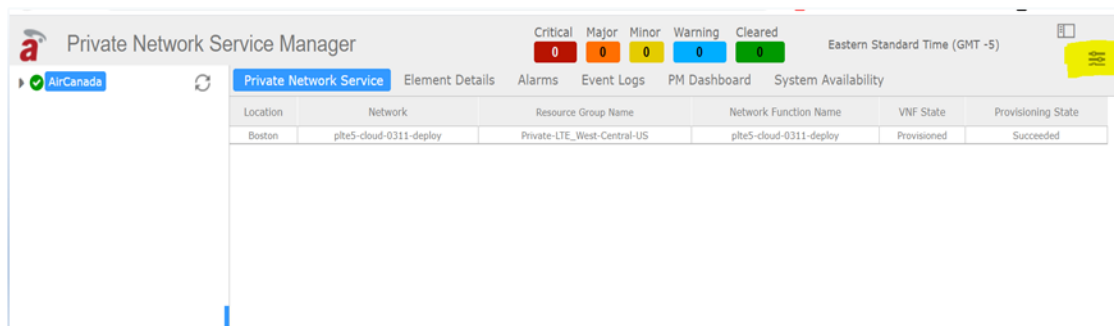
3. Enter the Private Network Service Manager credentials. The homepage appears.

Figure 3-10 Private Network Service Manager Login Page

4. Go to the top-right corner of the page, click the highlighted “Toggle Resource Selection View” icon . A new view appears.

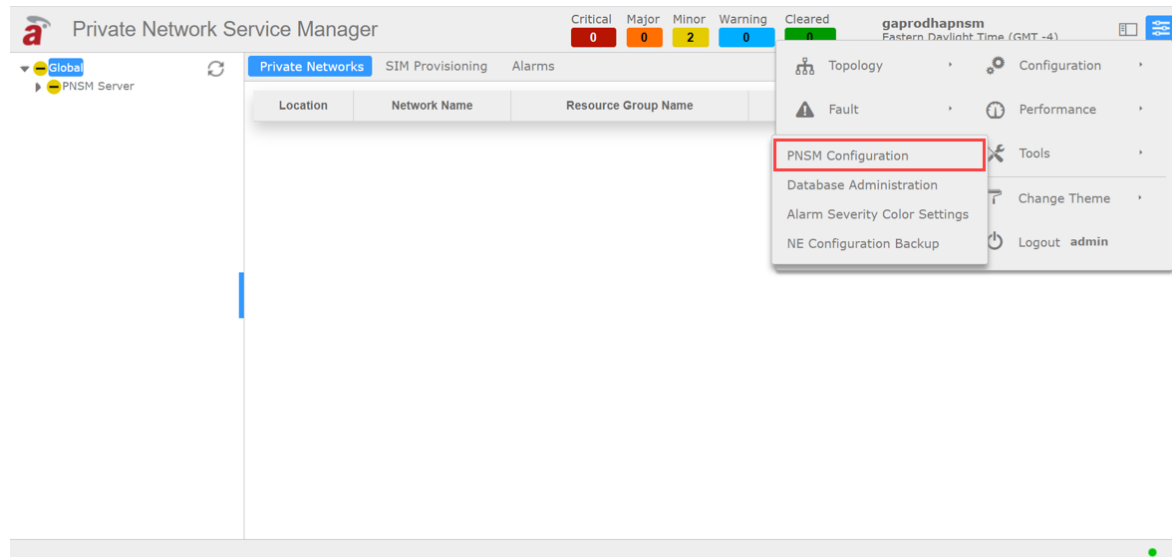
Figure 3-11 Toggling Resource Selection View

5. Go to the top-right corner of the page and click the  icon.

Figure 3-12 Selecting from the menu

6. From the displayed menu, click **Tools > PNSM Configuration**.

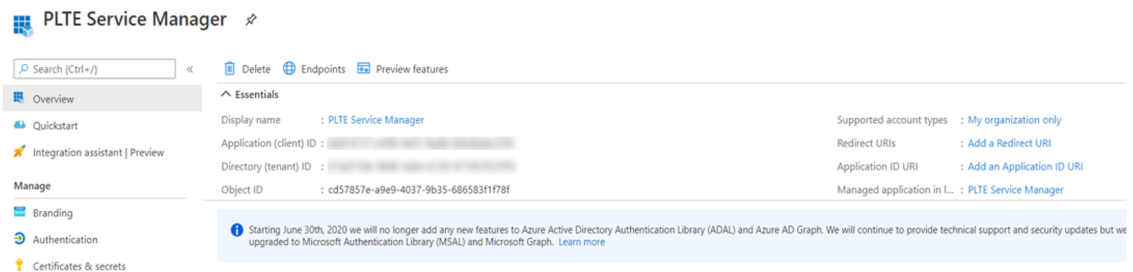
Figure 3-13 Configuring Service



7. The application must be registered with the Microsoft Identity Platform. To register the application, refer to [Registering an application with Microsoft Identity Platform](#).

Once the application is registered, the Application Registration Overview Page looks like the sample given below.

Figure 3-14 Application Registration Overview



8. Create a Custom Role. For more information on how to create a custom role in the Azure portal, refer to [Azure Custom Roles](#).

Figure 3-15 Creating Custom Role

Microsoft Azure (Preview) Search resources, services, and docs (G+/)

Home > Subscriptions > Private_LTE Dev >

Create a custom role

[Got feedback?](#)

Basics • Permissions • Assignable scopes JSON Review + create

To create a custom role for Azure resources, fill out some basic information. [Learn more](#)

* Custom role name

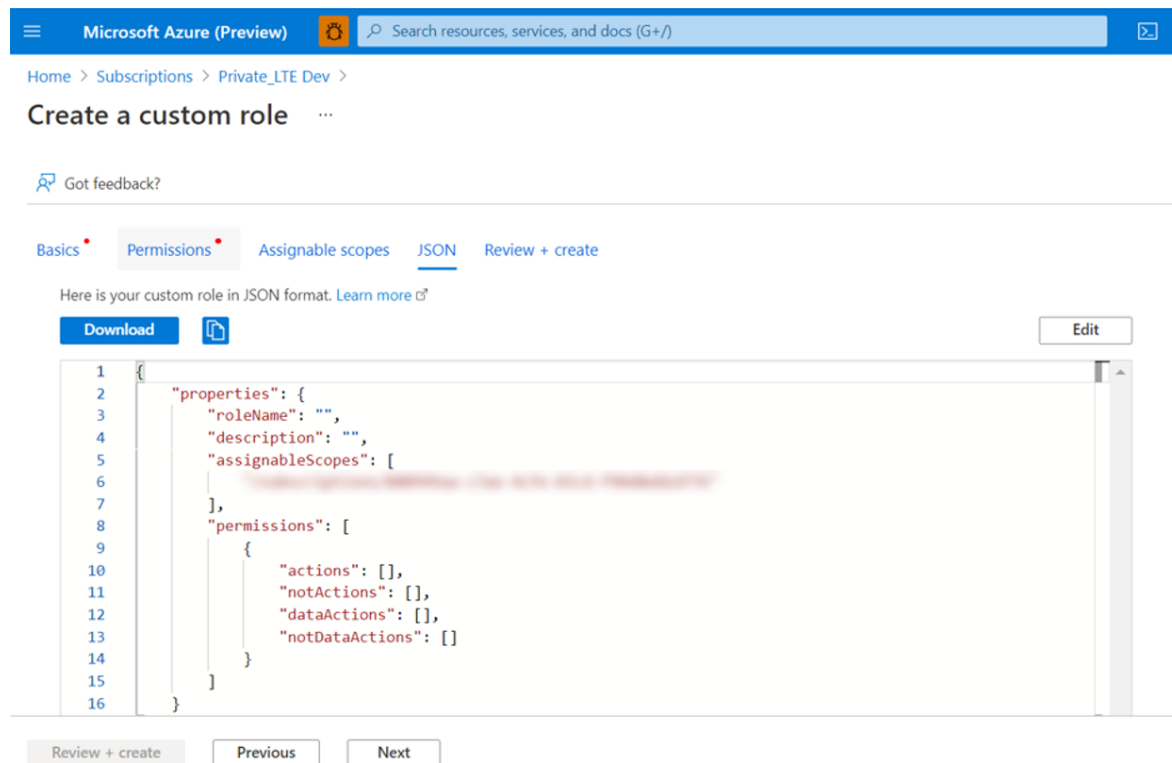
Description

Baseline permissions ☐ Clone a role ☒ Start from scratch ☐ Start from JSON

[Review + create](#) [Previous](#) [Next](#)

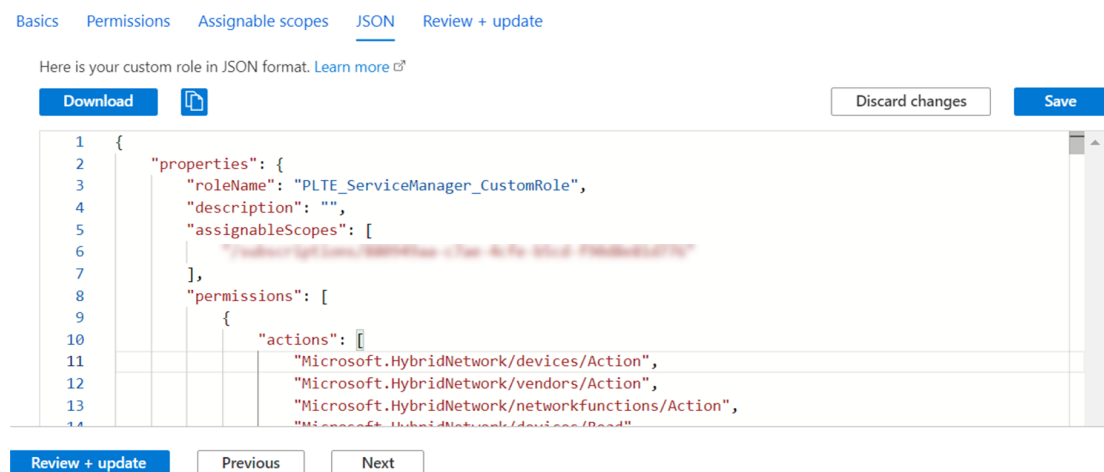
9. To assign all the required permissions, in the **Create a Custom Role** workflow page, select the **JSON** tab, Click **Edit** button visible on the top right corner of the JSON editor.

Figure 3-16 Adding Custom Role



- Open *azcustomrolepermissions.json* file in your system.
- Copy the permissions parameters from the .json file, replace the existing parameters with these updated parameters in the **permissions** section, click **Save**, and then click **Review+update**.

Figure 3-17 Replacing Custom Role Parameters



12. Assign the Custom Role to the registered application at Subscription level and note the **Application (Client) ID**, **Directory (tenant) ID**, and **Client Secret**.

13. Once the **Private Network Service Configuration Dialog** opens, configure the following attributes in UI with their respective values.
 - i. **Subscription ID:** Provide <Customer Subscription Id>
 - ii. **API Version:** 2020-01-01-preview
 - iii. **Vendor Name:** AffirmedVendor
 - iv. **Client ID:** Provide <Application (client) ID>. Use the Application (client) ID obtained in the previous step from the newly created application.
 - v. **Authority:** [https://login.microsoftonline.com/<Directory \(tenant\) ID>/oauth2/token](https://login.microsoftonline.com/<Directory (tenant) ID>/oauth2/token), where <Directory (tenant) ID> is obtained in the previous step from the newly created application
 - vi. **Client Secret:** Provide <Client Secret>. Use Client Secret obtained in the previous step from the newly created application.
14. The **ASAP server IP address** is auto-populated in the ASAP Server IP Address section. In case you want to reconfigure the IP address, provide the **ASAP server IP address**.
15. Select the **Provisioning Type** value. This value is set to **OP** by default and the default operator code is displayed.

***Note:** If the Provisioning Type is set to OPC, then the EPSSUBD OP value is not shown. The Provisioning Type is stored in the Service Manager database and you must provide the OPC value for each IMSI while importing SIMs.*

16. Edit the EPSSUBD OP value for the SIM Configuration as necessary.

***Note:** This value must be configured before any SIMs can be imported and activated.*

Figure 3-18 Azure Configuration Dialog

Private Network Service Configuration

ASAP Server

IP Address: 10.168.2.16

Azure Authentication Config

Subscription Id: [Redacted]

Client Id: [Redacted]

Authority: [Redacted]

Client Secret: [Redacted]

SIM Configuration

Provisioning Type: OP

EPSSUBD OP Value: CDC202D5123E20F62B6D676AC72CB318

Update

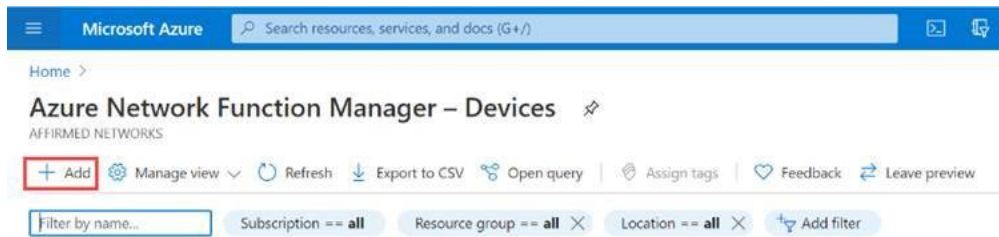
17. Click **Update**.

Adding Azure Network Function Manager – Device

You must create a device instance in Azure Network Function Manager-Device, which will be the digital twin of the Azure Stack Edge Device used by Azure Network Function Manager (ANFM).

To create a device instance:

1. Go to the [Azure Network Function Manager Devices](#) page in the Azure preview portal and click **Add**.

Figure 3-19 Adding Azure Network Function Manager – Device

2. Under the Basics tab:

- i. Select your **Resource group** from the dropdown.
- ii. Select **Region** (should be as per the resource group)
- iii. Provide **VM Name**
- iv. Select the correct **Azure Stack Edge** you would like to deploy the Netfoundry VM on.

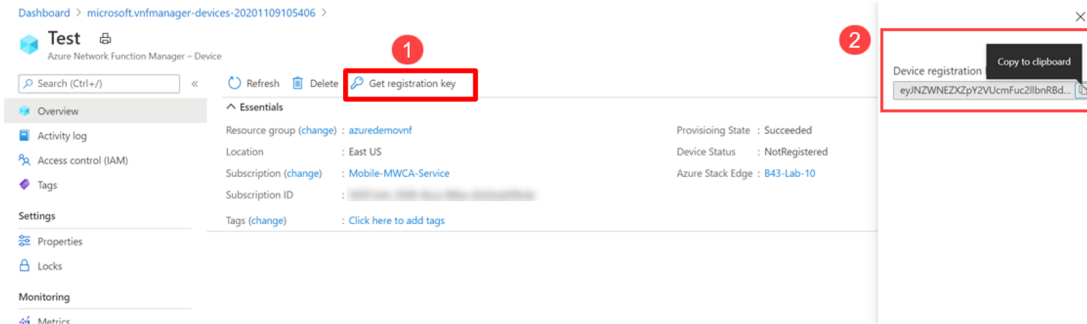
Figure 3-20 Creating Azure Network Function Manager Device

 The screenshot shows the 'Create Azure Network Function Manager - Device' form. At the top, there's a breadcrumb 'Home > Azure Network Function Manager – Devices >'. The title is 'Create Azure Network Function Manager - Device'. Below the title, there are tabs: 'Basics' (active), 'Tags', and 'Review + create'. Under the 'Basics' tab, there's a section 'Project details' with the instruction 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.' Below this, there are two dropdown menus: 'Subscription *' (selected: Mobile-MWCA-Service) and 'Resource group *' (empty, with a 'Create new' link below it). Below these is a section 'Instance details' with three fields: 'Region *' (selected: West Central US), 'Name *' (empty), and 'Azure Stack Edge *' (empty). At the bottom, there are three buttons: 'Review + create' (blue), '< Previous' (disabled), and 'Next : Tags >' (disabled).

3. Click **Review+create**.

- Once your device is provisioned, navigate to the resource group you deployed the device in, and click the Device resource to obtain the registration key by selecting **Get registration key**. Copy the Device registration key as it will be required in upcoming steps.

Figure 3-21 Getting Registration Key



- To register the device with the registration key obtained in the previous step, refer to [Connect to the Azure Stack Edge device via Windows PowerShell](#).
- Once PowerShell (minishell) is connected to the appliance, execute the following command with the device registration key you obtained in one of the previous steps as the parameter.

Invoke-MecRegister <device-registration-key>

Deploying NetFoundry Zero-Trust Network

Overview

NetFoundry v7.3 VMs is deployed in Azure cloud and Azure Stack Edge via [Azure Portal](#). Additionally, they will be registered against edge-routers and endpoints in the [NetFoundry web console](#), where services and policies (AppWANs) will be created to set up the zero-trust network. This network will be used for bi-directional connections between Private Network Service Manager in Azure cloud and EPC components (MME, MCC, and HSS) in Azure Stack Edge.

Note:

- Only one instance of NetFoundry router is deployed in each ASE box.
- Only one instance of NetFoundry will suffice in the Azure cloud.

Creating Fabric Routers

Fabric routers are deployed per network and they are NetFoundry-hosted edge routers.

To create fabric edge routers:

- Provide Router name.

Note: Two routers are created for the HA Purpose. Router names: `fabric_router_01`, `fabric_router_02`.

2. Select the Azure location based on the requirements and click the **Create** button.

Figure 3-22 Router Names

Prerequisites

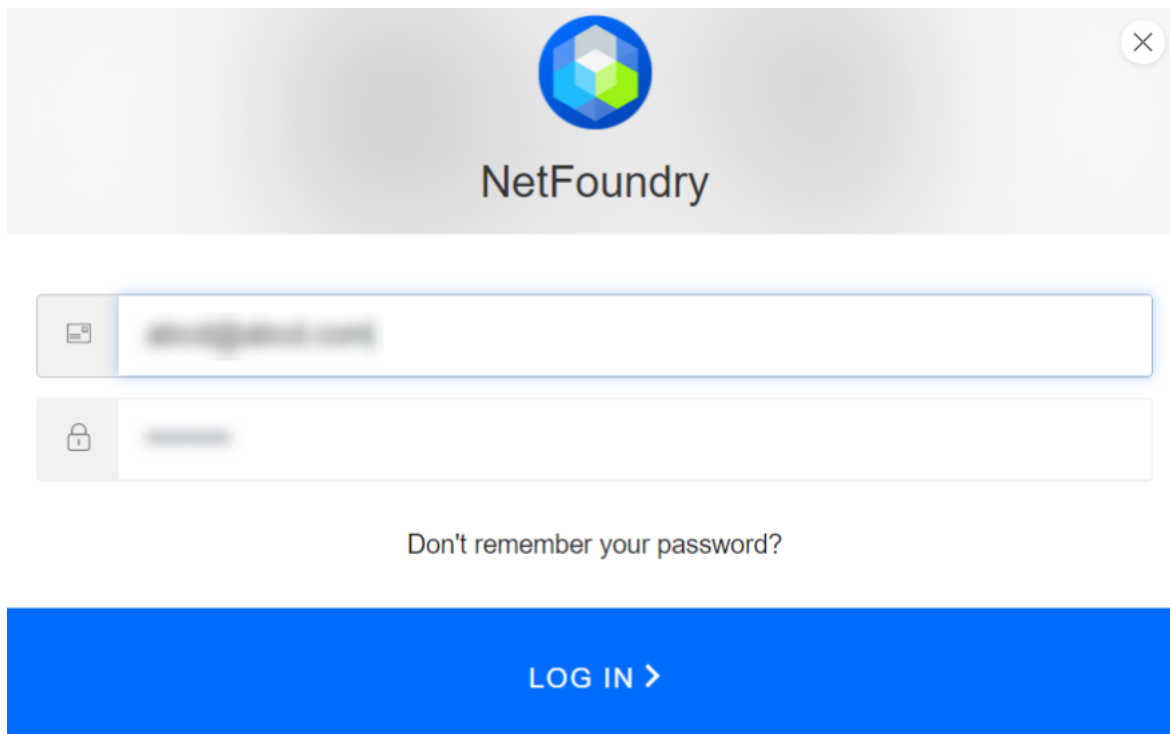
- Edge Routers and Endpoints must be created before deploying NetFoundry VMs to generate the registration keys for NetFoundry VMs.
 - Logical Edge Router resources must be created in the NetFoundry console so that the VMs deployed in Azure and ASE can be connected to the NetFoundry console using router registration keys to register with the NetFoundry controller.

Creating Edge Router

To create an Edge Router:

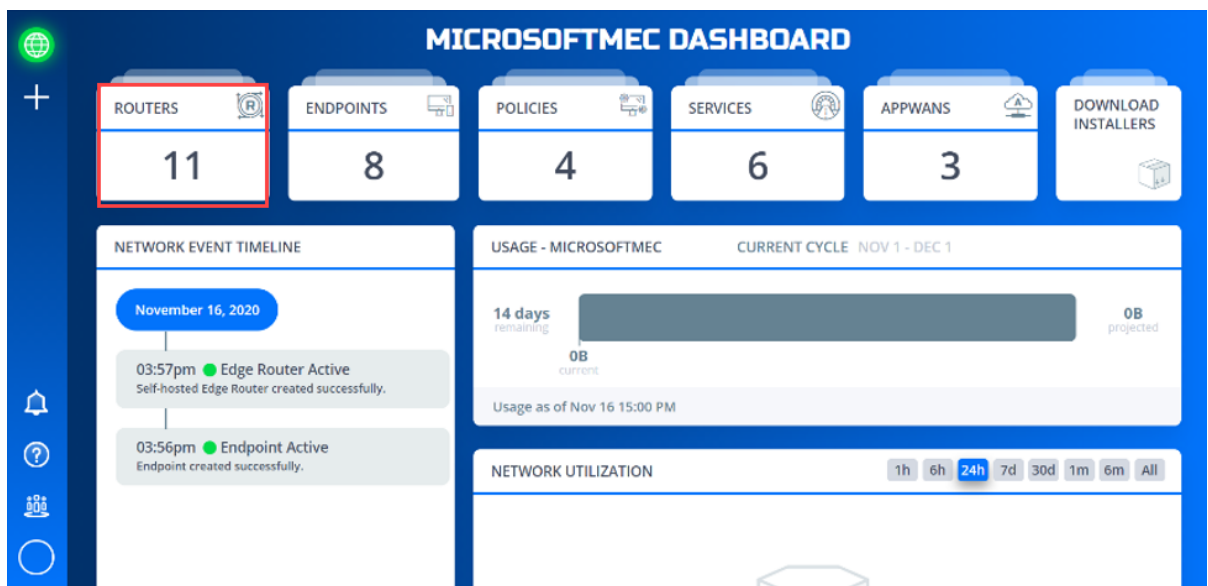
1. Enter your NetFoundry credentials and **Log In**.

Figure 3-23 NetFoundry Login Page



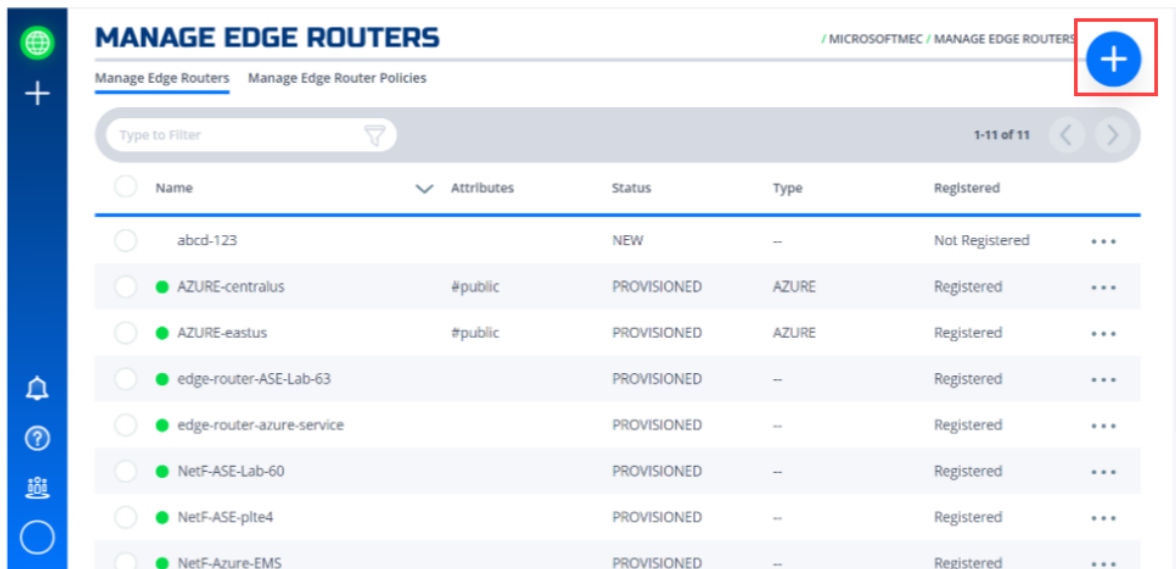
2. Once the homepage opens, go to the top of the page and click **ROUTERS**.

Figure 3-24 Clicking Routers



3. Click (+) to create a new edge router.

Figure 3-25 Creating a New Edge Router



4. Provide a name for the new edge router for NetFoundry VM in Azure Cloud. Select the Datacenter from the **DATACENTER** dropdown according to the region and click **Create**.

Figure 3-26 Providing Router Attributes

CREATE A NEW EDGE ROUTER
Enter your routers details
[What's a Edge Router?](#)

1 ROUTER NAME REQUIRED

test-1

2 ROUTER ATTRIBUTES ⓘ OPTIONAL

Add attributes for grouping routers

✕ #test-1

3 SELECT A HOSTING TYPE REQUIRED

☒ CUSTOMER ⓘ
HOSTED

LINK LISTENER OPTIONAL

ENABLE LINK LISTENER ⓘ

☐ OFF

☐ NETFOUNDRY ⓘ
HOSTED

DATACENTER REQUIRED

Select a Datacenter

CREATE

5. To copy the Registration Key, click the **Registration Key** button. This registration key will be used in Router Registration property in **Custom Data** in Azure.

Figure 3-27 Copying Registration Key

CONGRATULATIONS

Your Edge Router has been created

Register your router in two simple steps.

1 Copy Your Registration Key

ABCD-1234 REGISTRATION KEY TAP TO COPY

REGISTRATION KEY

ADVANCED DOWNLOADS

ADVANCED DOWNLOADS

SUCCESS

Reg Key Copied
The reg key [REDACTED] has been copied to your clipboard

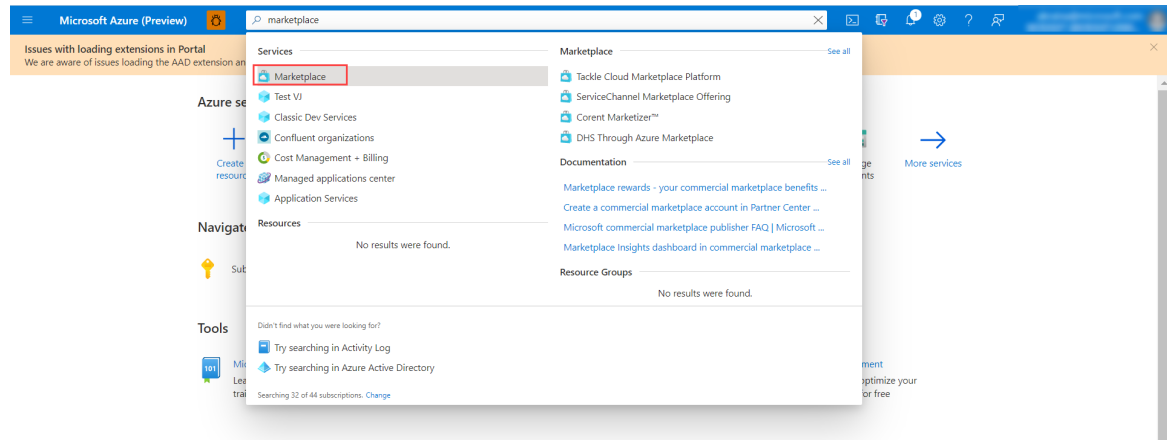
- Repeat the above two steps to create a second edge router and download the registration key for NetFoundry VM in Azure Stack Edge.

Deploying NetFoundry VM in Azure Cloud

To deploy NetFoundry VM in Azure Cloud:

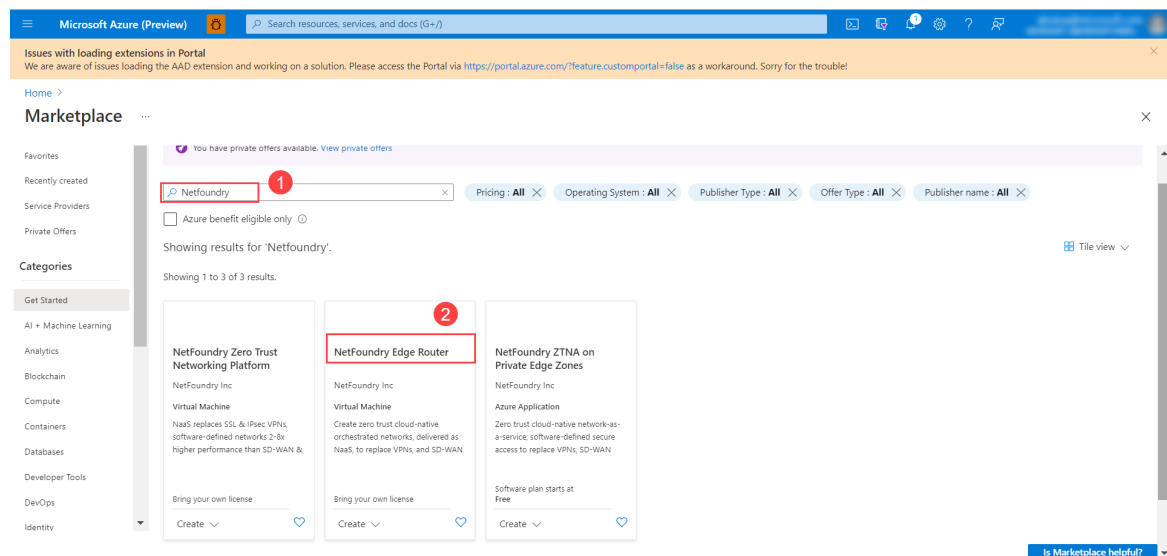
- Log in to the [Azure portal](#), search for "Marketplace" in the search bar and select Marketplace from the dropdown. The Marketplace page appears.

Figure 3-28 Selecting Marketplace



- Type "Netfoundry" in the search bar, and click **NetFoundry Edge Router**.

Figure 3-29 Selecting NetFoundry Edge Router



- Click **Create**.

Figure 3-30 Creating NetFoundry Edge Router

Home > NetFoundry Edge Router  ...

NetFoundry Inc

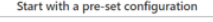


NetFoundry Edge Router  Add to Favorites

NetFoundry Inc

★ 0.0 (0 Azure ratings) | ★ 0.0 (0 external ratings)

Subscription

AISC-DEV-02   

Want to deploy programmatically? [Get started](#)

Overview Plans + Pricing Usage Information + Support Reviews

NetFoundry Edge Routers provide zero trust optimized connectivity between Azure Cloud and any site, edge device, private/public clouds, and hybrid applications.

Edge Routers are either NetFoundry hosted in the NetFoundry global fabric or Customer hosted. Edge Routers provide ingress/egress of network traffic between endpoints and clouds. NetFoundry has pre-integrated our Edge Routers with Azure. This allows for rapid deployment of zero trust, high performance programmable networking at scale.

NetFoundry eliminates the need for expensive private circuits, proprietary hardware, and traditional VPNs. Use NetFoundry's open SDKs and APIs to embed programmable networking into any app, host, device, or virtual cloud network.

Like what Azure does for infrastructure, NetFoundry implements a fully managed Network-as-a-Service solution. You have full control of your networks without having to manage the underlying infrastructure. Cloud-native orchestration and automation abstracts away complexity and simplifies management.

NetFoundry Networks Deliver:

- Global, high performance, zero trust networks, with edge router integrations with Azure
- NaaS provides real-time, dynamic routing optimization to minimize latency and packet loss
- Spin up new networks in minutes with cloud-native orchestration
- Integrated Secure Web Gateway delivers secure access service edge (SASE) services
- Zero trust connectivity for data transport, remote management, and segmented vendor access
- Rich APIs for integrations with DevOps tools, and management services
- Software enforced; policy driven access e.g. only open up network connections with specific services at specific times
- All infrastructure managed by NetFoundry as-a-service

Business Outcomes:

- Accelerate agile digital business processes with secure, flexible, dynamic networking that can deliver any app to any user anywhere
- Simply and securely extend your network to Azure from anywhere with zero trust and SASE capabilities
- Strengthen business continuity processes with a cloud orchestrated global Network-as-a Service platform that rapidly scales on-demand as business and user requirements change
- Drive continuous cost optimization with exceptional quality of experience
- Simplify by eliminating the headaches and complexities of MPLS and VPNs

4. Once **Create a Virtual Machine** page opens, under the **Basics** tab:
 - i. use the same **Subscription** and **Resource group** as the one for Private Network Service Manager (**PLTE_Service_Manager** in the previous example)
 - ii. Region must be in accordance with where the resource group resides.

Figure 3-31 Creating a Virtual Machine

Microsoft Azure (Preview)  Search resources, services, and docs (5+)

Home > Marketplace > NetFoundry Edge Router >

Create a virtual machine ...

Basics Disks Networking Management Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription  Private_LTE Dev

Resource group  (New) NFSMCLLOUD_group 

Instance details

Virtual machine name  NFSMCLLOUD

Region  (US) East US

Availability options  Availability zone

Availability zone  1

Image  Ziti Edge Router - Gen1 

Azure Spot instance 

- Go to the **Networking** tab, select the **Virtual network** and **Subnets** from the respective dropdowns.

Figure 3-32 Selecting Virtual Machine and Subnet

Microsoft Azure (Preview) Report a bug Search resources, services, and docs (G+/I)

Home > Marketplace > NetFoundry Edge Router >

Create a virtual machine

Basics Disks **Networking** Management Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * [Create new](#)

Subnet * [Manage subnet configuration](#)

Public IP [Create new](#)

NIC network security group ☐ None ☐ Basic ☒ Advanced

Info This VM image has preconfigured NSG rules

Info The selected subnet 'gspite-vnet-mgmt (10.60.0.0/24)' is already associated to a network security group 'gspite-vnet-gspite-vnet-mgmt-NSG'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

Configure network security group *

[Review + create](#) < Previous Next: Management >

- Go to the **Advanced** tab and fill the **Custom data**.

Figure 3-33 Filling Custom Data

Microsoft Azure (Preview) Report a bug Search resources, services, and docs (G+/I)

Home > Marketplace > NetFoundry Edge Router >

Create a virtual machine

Basics Disks Networking Management **Advanced** Tags Review + create

Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions [Select an extension to install](#)

Custom data

Pass a script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

Info Your image must have a code to support consumption of custom data. If your image supports cloud-init, custom-data will be processed by cloud-init. [Learn more about custom data for VMs](#)

User data

Pass a script, configuration file, or other data that will be accessible to your applications **throughout the lifetime of the virtual machine**. Don't use user data for storing your secrets or passwords. [Learn more about user data for VMs](#)

Enable user data ☐

[Review + create](#) < Previous Next: Tags >

- Paste the below script in custom data section. Replace Router_Registration and Endpoint_Key with the keys you download from NetFoundry web console in the next section

```
#!/bin/bash -xe
```

```
#Start logging
```

```
exec >>(tee /var/log/user-data.log|logger -t user-data -s  
2>/dev/console) 2>&1
```

```
ROUTER_REGISTRATION="PLACE_ROUTER_REGISTRATION_HERE"
```

```
/opt/netfoundry/router-registration ${ROUTER_REGISTRATION}
```

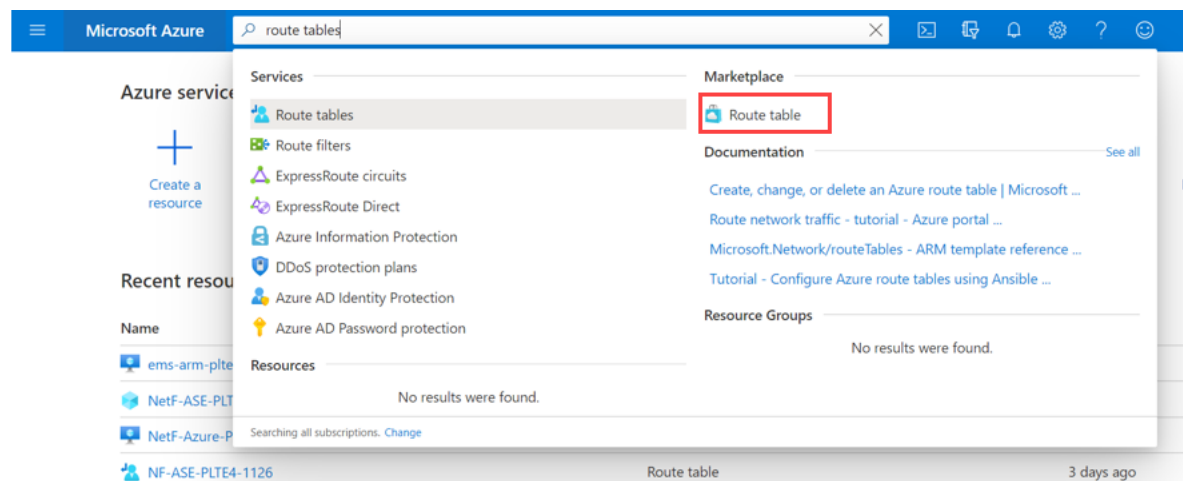
8. Click **Review+create**.

Creating a Route Table

To create a Route Table:

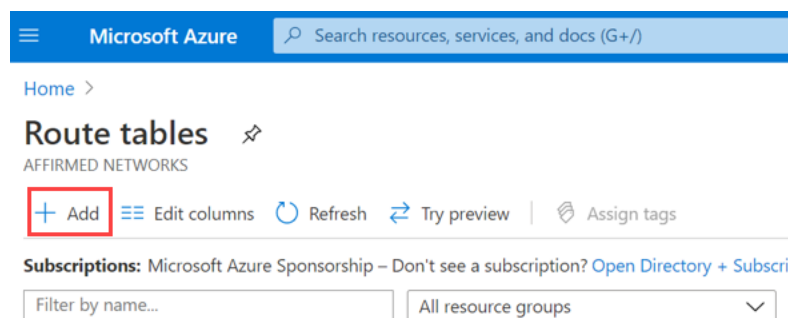
1. Log in to the [Azure portal](#), search for "Route table" in the top search bar, and click **Route Tables** appearing under **Marketplace**.

Figure 3-34 Selecting Route Tables



2. Once the Route Tables page opens, click the **+Add** button.

Figure 3-35 Adding the Route Table



3. Provide **Resource Group** name and select **Region** (both the values must be same as for Private Network Service Manager and NetFoundry VM in Azure), and then click **Review+create**.

Figure 3-36 Validated Information for Route Table

Microsoft Azure Search resources, services, and docs (G+)

Home > Route tables >

Create Route table

Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Microsoft Azure Sponsorship

Resource group * ⓘ Private_LTE_EMS
[Create new](#)

Instance details

Region * ⓘ East US

Name * ⓘ Service-Manager-To-ASE ✓

Propagate gateway routes * ⓘ ☒ Yes ☐ No

Review + create < Previous Next : Tags >

4. Once the information is validated, click **Create**.

Figure 3-37 Creating a Route Table

Microsoft Azure Search resources, services, and docs (G+)

Home > Route tables >

Create Route table

✓ Validation Passed

Basics Tags **Review + create**

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

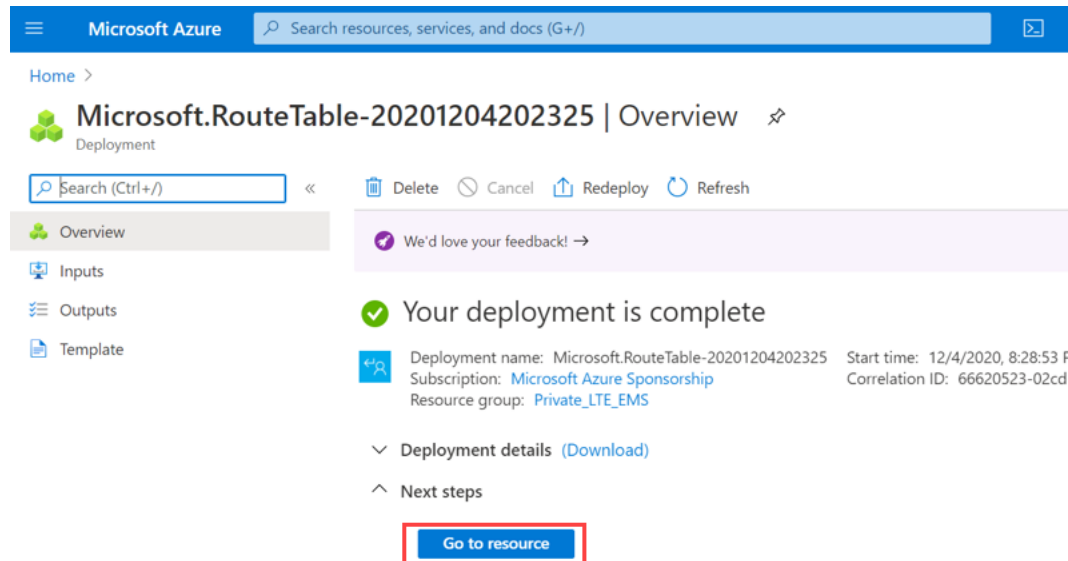
Basics

Subscription	Microsoft Azure Sponsorship
Resource group	Private_LTE_EMS

Create < Previous Next [Download a template for automation](#)

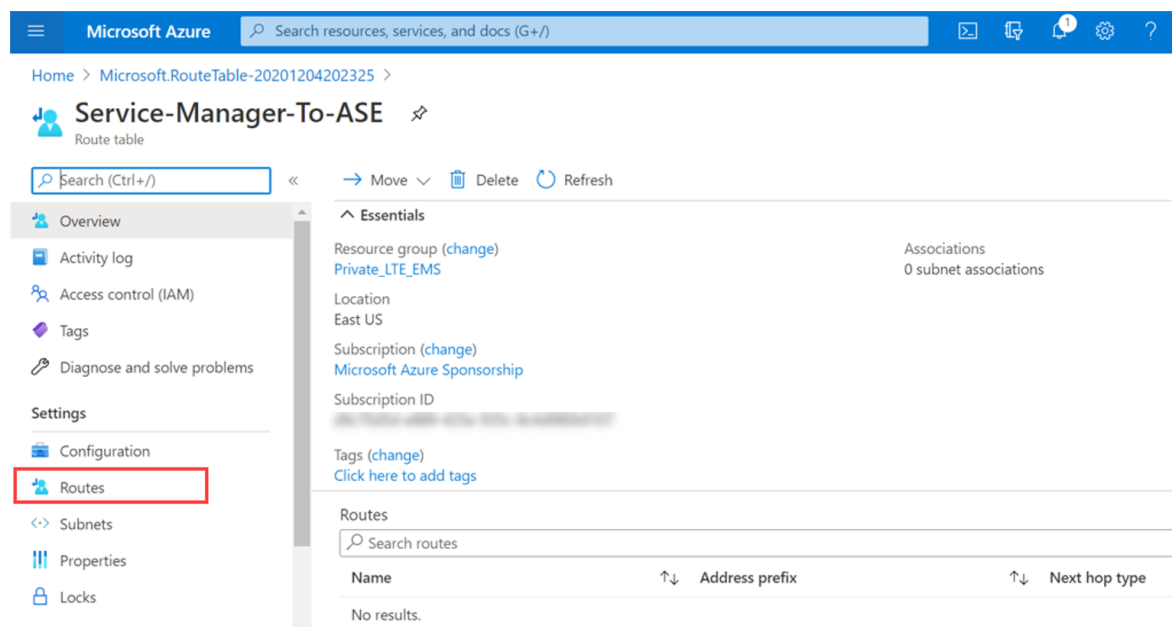
5. Once Route Table deployment is complete, click **Go to resource**.

Figure 3-38 Route Table Deployment Completion

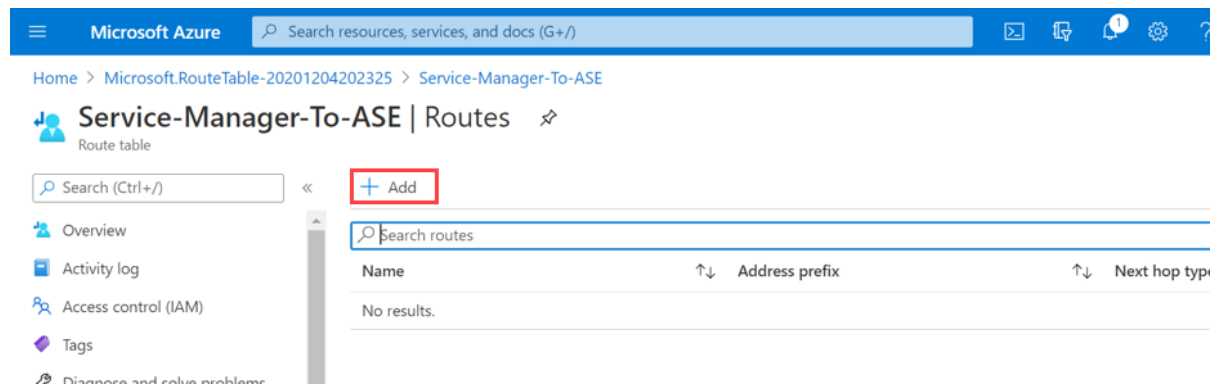


6. Once the page opens, click **Routes**.

Figure 3-39 Selecting Routes



7. Once the page opens, click **+Add**.

Figure 3-40 Adding Route

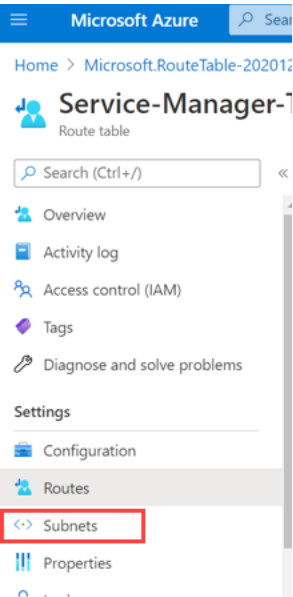
8. Provide the following information:

- i. **Route Name:** Name of the Route
- ii. **Address prefix:** Subnet assigned to Azure Stack Edge Box
- iii. **Next hop type:** Virtual appliance
- iv. **Next hop address:** Management IP address of NetFoundry VM created in previous section
- v. Click **OK**.

Figure 3-41 Providing Router Information

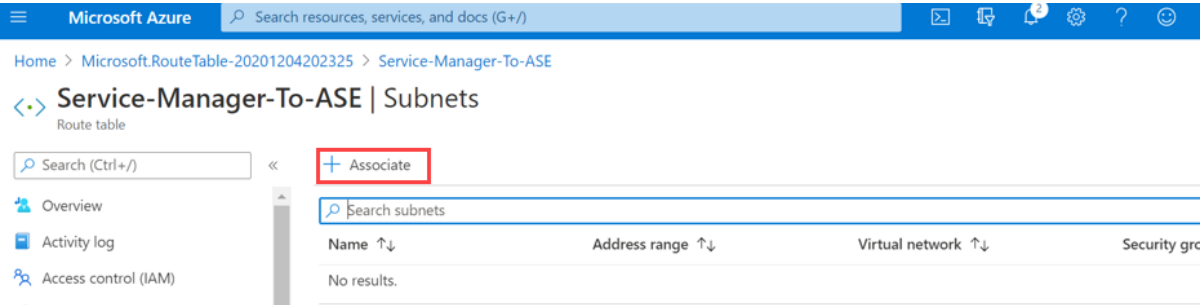
9. Once the route is created, click **Subnets**.

Figure 3-42 Selecting Subnets



10. Click +Associate.

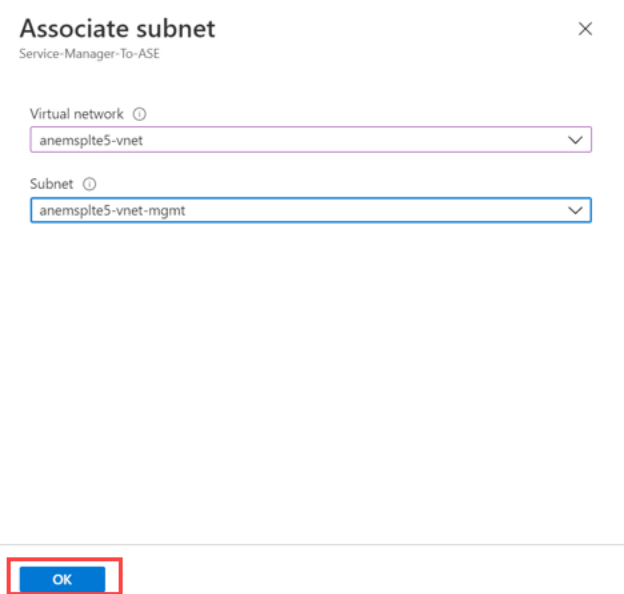
Figure 3-43 Selecting Associate



11. Provide the following values:

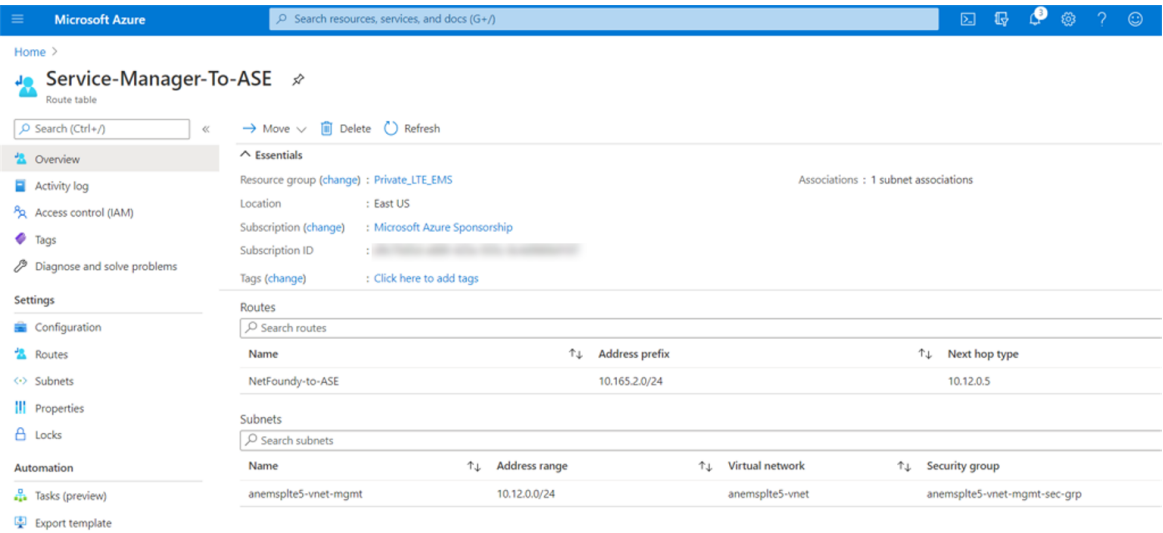
- i. **Virtual Network:** VNet of Private Network Service Manager/ NetFoundry VM
- ii. **Subnet:** Management Subnet of Private Network Service Manager/ NetFoundry VM
- iii. Click **OK**.

Figure 3-44 Providing Subnets



12. Once the subnet is created, click **Overview**. The route table should look like this.

Figure 3-45 Route Table Overview

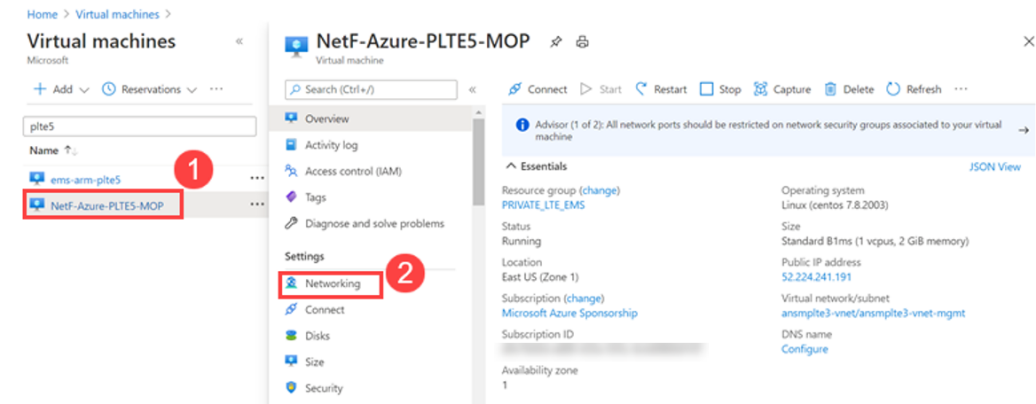


Enabling IP forwarding for NetFoundry VM in Azure

To enable IP forwarding for NetFoundry VM in Azure:

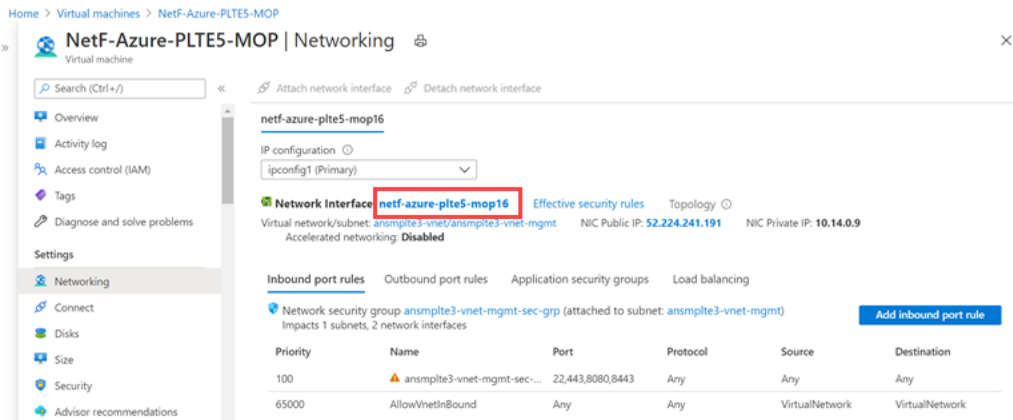
1. Select the NetFoundry VM and click **Networking**.

Figure 3-46 Selecting the NetFoundry VM



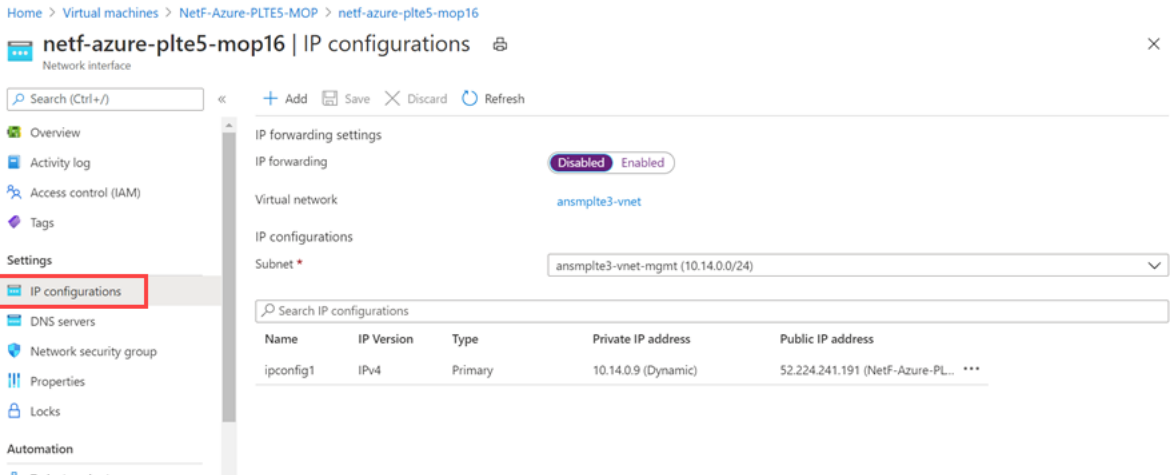
2. Click the **Network Interface** name.

Figure 3-47 Selecting Network Interface



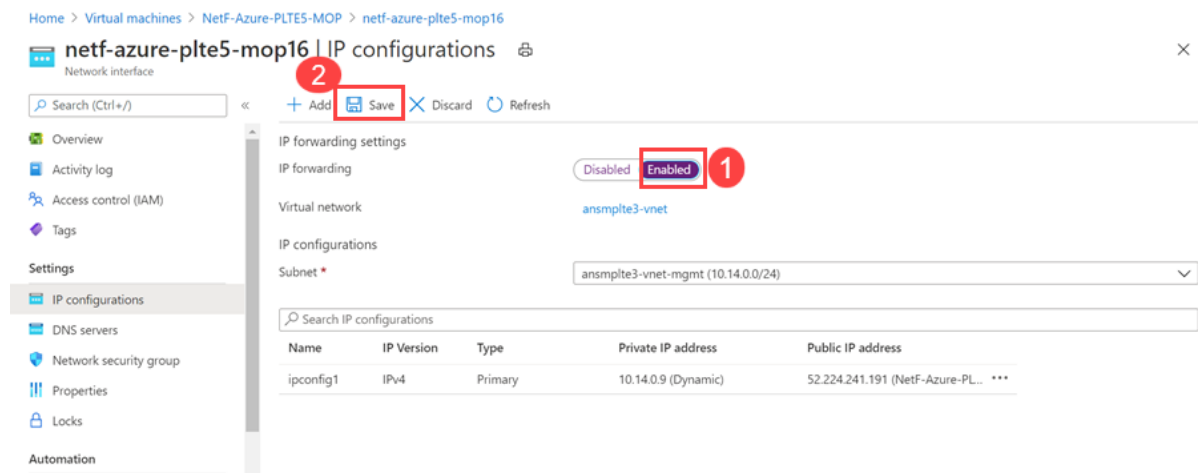
3. Click **IP configurations**

Figure 3-48 Selecting IP Configurations



4. Toggle **IP forwarding** to **Enabled** and click **Save**.

Figure 3-49 Enabling the IP Forwarding



NetFoundry VM Deployment in Azure Stack Edge

Prerequisites

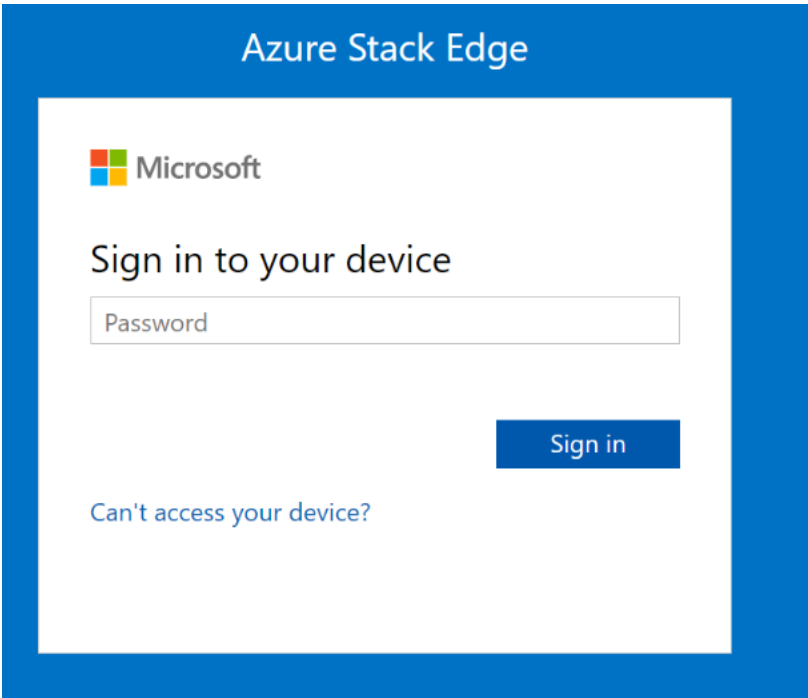
Note: DNS IP address must be populated in the ASE box for the port in which Compute is enabled.

- Provide DNS Server

To provide DNS Server:

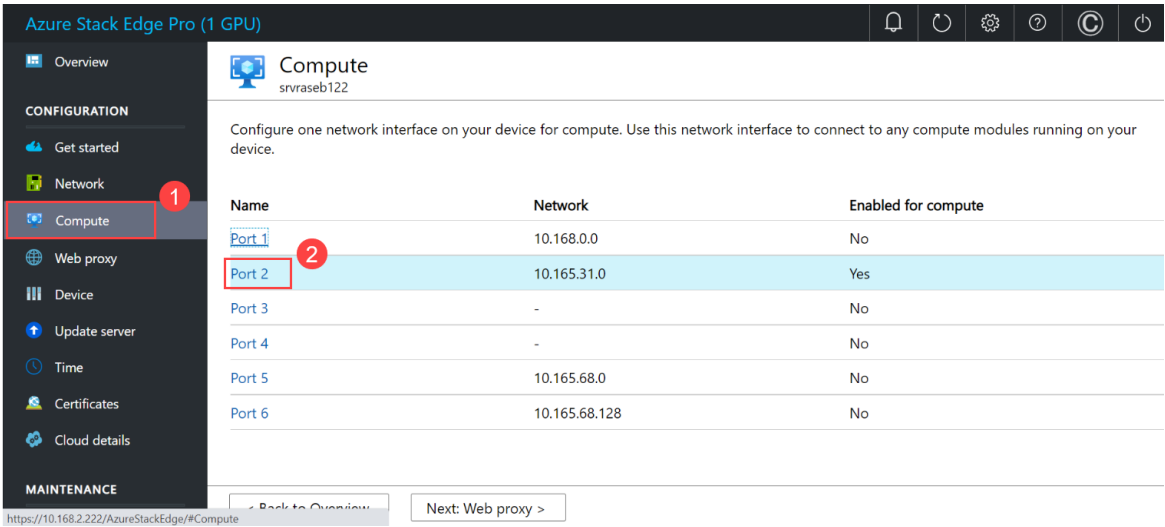
1. Go to ASE box Management IP Address and provide password.

Figure 3-50 ASE Box Login



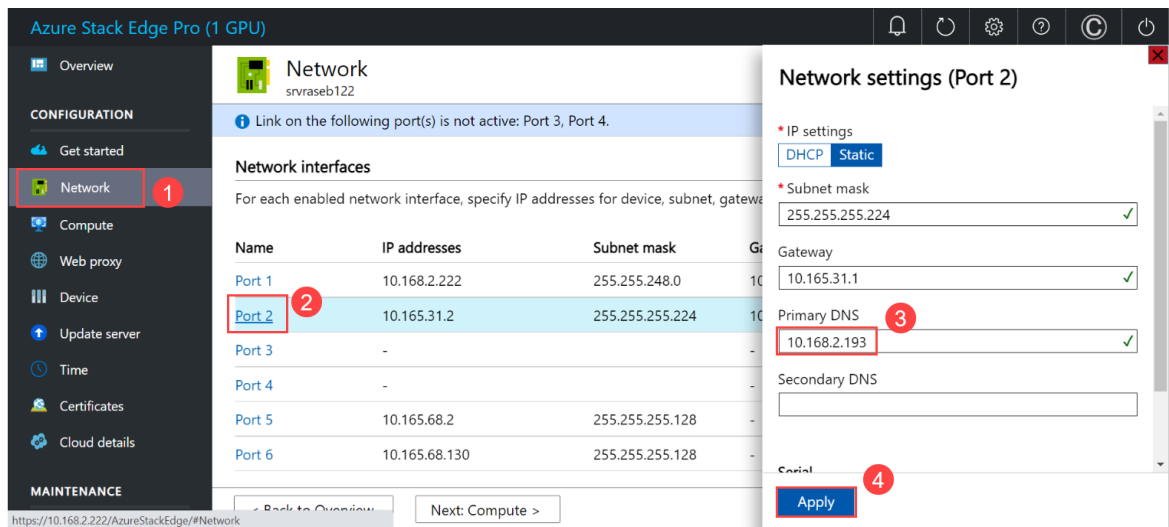
2. Under the **Configuration** section, click **Compute**, and then select the port for which **Enabled for compute** is set to **Yes**. For example, Port 2.

Figure 3-51 Selecting the Port



3. Under the **Network** section, click the respective port, provide the **Primary DNS** address, and then click **Apply**.

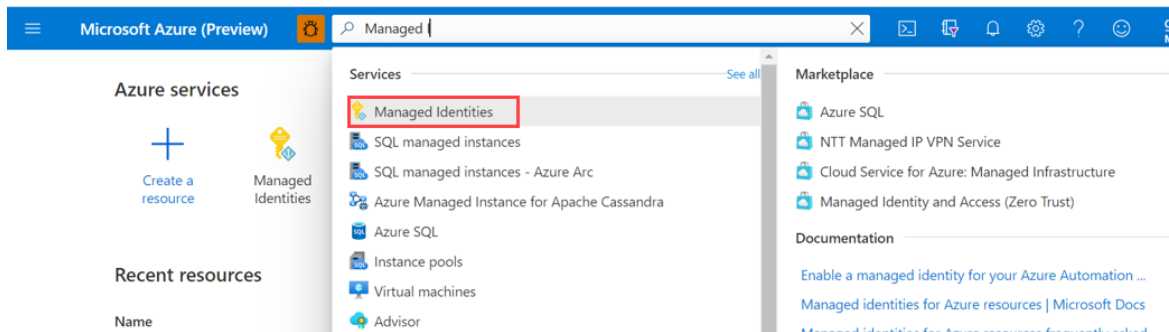
Figure 3-52 Providing DNS Server



- To create Managed Identities:

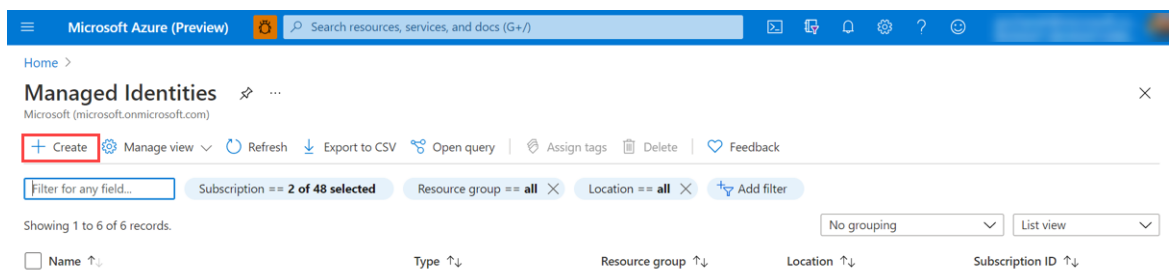
1. Go to the search bar, type "Managed Identities" and select **Managed Identities** from the dropdown. The Managed Identities page opens.

Figure 3-53 Selecting Managed Identities



2. Click **Create**. The **Create User Assigned Managed Identity** page appears.

Figure 3-54 Creating Managed Identities



3. Select **Subscription**, **Resource Group**, **Region** from their respective dropdowns, provide **Name**, and click **Review+Create**.

Figure 3-55 Providing Managed Identities Details

Microsoft Azure (Preview)

Search resources, services, and docs (G+ /)

[Home](#) > [Managed Identities](#) >

Create User Assigned Managed Identity ...

Basics

Tags

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Private_LTE Dev

Resource group * ⓘ

Private_LTE_ASE

Create new

Instance details

Region * ⓘ

East US

Name * ⓘ

Review + create

< Previous

Next : Tags >

4. Verify the details and click **Create**.

Figure 3-56 Creating Managed Identity

Microsoft Azure (Preview) Search resources, services, and docs (G+/)

Home > Managed Identities >

Create User Assigned Managed Identity ...

✓ Validation Passed

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Basics

Subscription	Private_LTE Dev
Resource group	Private_LTE_ASE
Region	East US
Name	test

[Create](#)
[< Previous](#)
[Next](#)
[Download a template for automation](#)

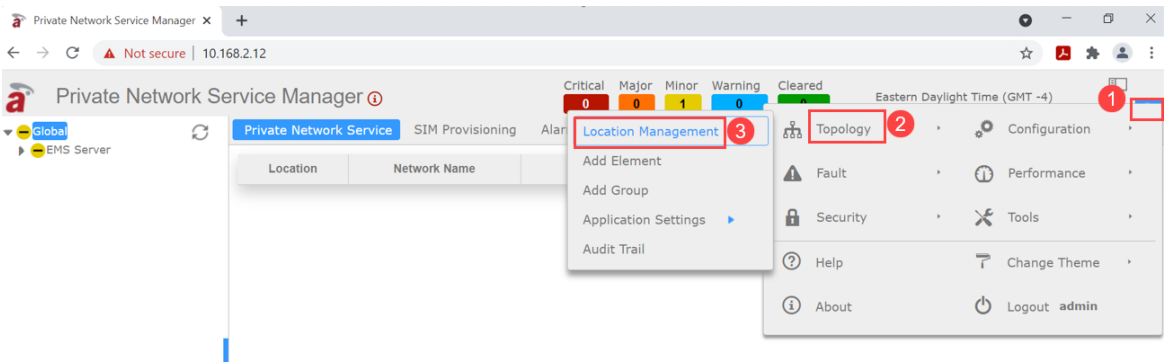
5. Associate a Custom Role to this newly created Network Identity.

Deploying NetFoundry VM in Azure Stack Edge

To deploy NetFoundry VM in Azure Stack Edge:

1. Log in to Private Network Service Manager.
2. Click main menu, select **Topology**, and then click **Location Management**.

Figure 3-57 Adding Location Management



3. Once the Location Management dialog appears, provide **Location Name**, **Management Subnet**, **LAN Subnet**, and **WAN Subnet**, and then click **Submit**.

Figure 3-58 Providing Location Details

The screenshot shows the 'Location Management' dialog box. On the left, there's a 'Locations' list with 'Richardson' and 'CUPS_LOCATION'. On the right, there's an 'Add Location' form with the following fields: 'Location Name', 'Management Subnet', 'Management Gateway', 'LAN Subnet', 'LAN Gateway', 'WAN Subnet', and 'WAN Gateway'. Each field has a corresponding input box. At the bottom right, there is a 'Submit' button.

4. Once the location is created, two keys namely, public and private keys can be used with the VNF. Download **Public Key** using the **Download Public Key** button for future use, and close the dialog.

Figure 3-59 Downloading Public Key

The screenshot shows a web application window titled "Location Management". On the left, there is a "Locations" list with two entries: "Richardson" (highlighted with a blue border and a plus icon) and "CUPS_LOCATION" (with a trash icon). On the right, the "Location Details" form is displayed for the "Richardson" location. The form includes the following fields and actions:

Field/Action	Value/Icon
Location Name	Richardson
Management Subnet	10.165.32.0/22
Management Gateway	
LAN Subnet	10.165.60.0/22
LAN Gateway	
WAN Subnet	10.165.64.0/23
WAN Gateway	
Download Public Key	
Download Private Key	
Next Available IP Addresses	

A "Submit" button is located at the bottom right of the form.

5. Retrieve IP Address for the NetFoundry VM from Location Management page. To do that, refer to *Creating a Location* section in *Private Network Service Manager User's Guide*.
6. Use the following link to get into Azure portal NetFoundry Market Place offer: [NetFoundry ZTNA on Private Edge Zones](#)
7. Once you fill and submit the contact details, you will be directed to the **NetFoundry Edge Router** page.

Figure 3-60 NetFoundry Edge Router Page

[Home](#) > [Marketplace](#) >

NetFoundry ZTNA on Azure private MEC

NetFoundry Inc



NetFoundry ZTNA on Azure private MEC

NetFoundry Inc

Plan

Application Ziti Edge Router

Create

[Add to Favorites](#)

Overview

Plans + Pricing

Usage Information + Support

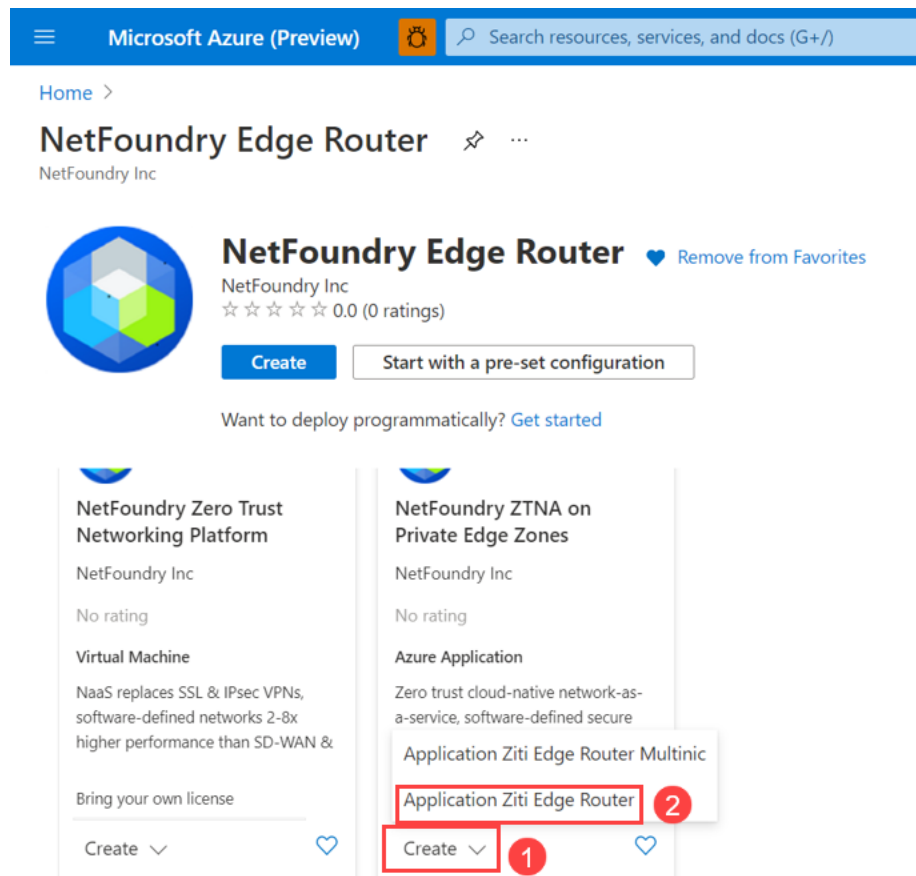
Reviews

NetFoundry ZTNA provides simple, zero trust optimized connectivity between Azure Stack Edge and any site, edge device, cloud, private, public and hybrid applications.

NetFoundry's integration with Azure Stack Edge enables you to instantly spin up zero trust, high performance, application-specific, global networks at scale using NetFoundry's cloud native orchestration tools and APIs. It eliminates the need for expensive private circuits, proprietary hardware, and traditional VPNs. Like what Azure Stack Edge does for infrastructure (Hardware-as-a-Service), NetFoundry implements a fully managed Network-as-a-Service (NaaS) solution, empowering you to orchestrate and control your networks without managing the underlying infrastructure. Combine Stack Edge with ZTNA to create seamless edge-to-cloud connectivity for any workload or edge use case including Azure Edge Zones.

NetFoundry ZTNA Delivers

- Under the **NetFoundry ZTNA on Private Edge Zones** section, click **Create**, and then select **Application Ziti Edge Router**.

Figure 3-61 Selecting Application Ziti Edge Router

9. Under the **Basics** tab, select **Resource Group**, select **Azure VNF Manager Device**, provide **NetFoundry VNF Name**, **Edge Router Registration Key**, **Endpoint JWT**, **Public SSH key**, and **Application Name**, and click **Next**.

Note: Refer the Prerequisites section for Router Registration Key and Endpoint Key.

Note: Refer step-4 to use Public SSH Key.

Figure 3-62 Providing Router Information

Home > Marketplace >

Create NetFoundry ZTNA on Azure private MEC ...

1 Basics

2 Network settings

3 Managed Identity Details

4 Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Private_LTE Dev

Resource group *

Create new

Instance details

Region *

East US

Azure VNF Manager device

ase-plte-10

NetFoundry VNF Name *

NFASEPLTE10

Edge Router Registration Key *

Public SSH Key

Managed Application Details

Provide a name for your managed application, and its managed resource group. Your application's managed resource group holds all the resources that are required by the managed application which the consumer has limited access to.

Application Name *

NFASEPLTE10APP

Managed Resource Group *

mrg-application-ziti-private-edge-20210823175845

< Previous

Next

10. Under the **Network Settings** section, provide **IP address** for NetFoundry VM, and its corresponding **Subnet**, **Default Gateway**, and **DNS IP Address**, and then click **Next**.

***Note:** The NetFoundry Edge Router can only be deployed on the Management Network.*

Figure 3-63 Network Setting tab

✓ Basics
2 Network settings
3 Review + create

Management

IP address * ⓘ

Subnet * ⓘ

Default gateway * ⓘ

DNS 1

DNS 2

- Under the **Managed Identity Details** tab, click on **Add** button to add a **Managed Identity**, and select the **Subscription** from the dropdown, select the **managed identity**, and then click **Add**. The identity is added.

Figure 3-64 Providing Managed Identity Details

Microsoft Azure (Preview) Search resources, services, and docs (G+/I)

Home > Marketplace >

Create NetFoundry ZTNA on Private Edge Zones ...

✓ Basics
✓ Network settings
1 Managed Identity Details
4 Review + create

User assigned managed identity

Add user assigned identities to grant the managed application access to Azure Network Function Managed device.

+ Add 1 Remove

Name	resource group	subscription
No user assigned managed identities assigned to this resource. Select 'Add' to add more.		

< Previous Next

Add 4

Subscription * 2 Private_LTE Dev

User assigned managed identities

Filter by identity n 3 and/or resource group name

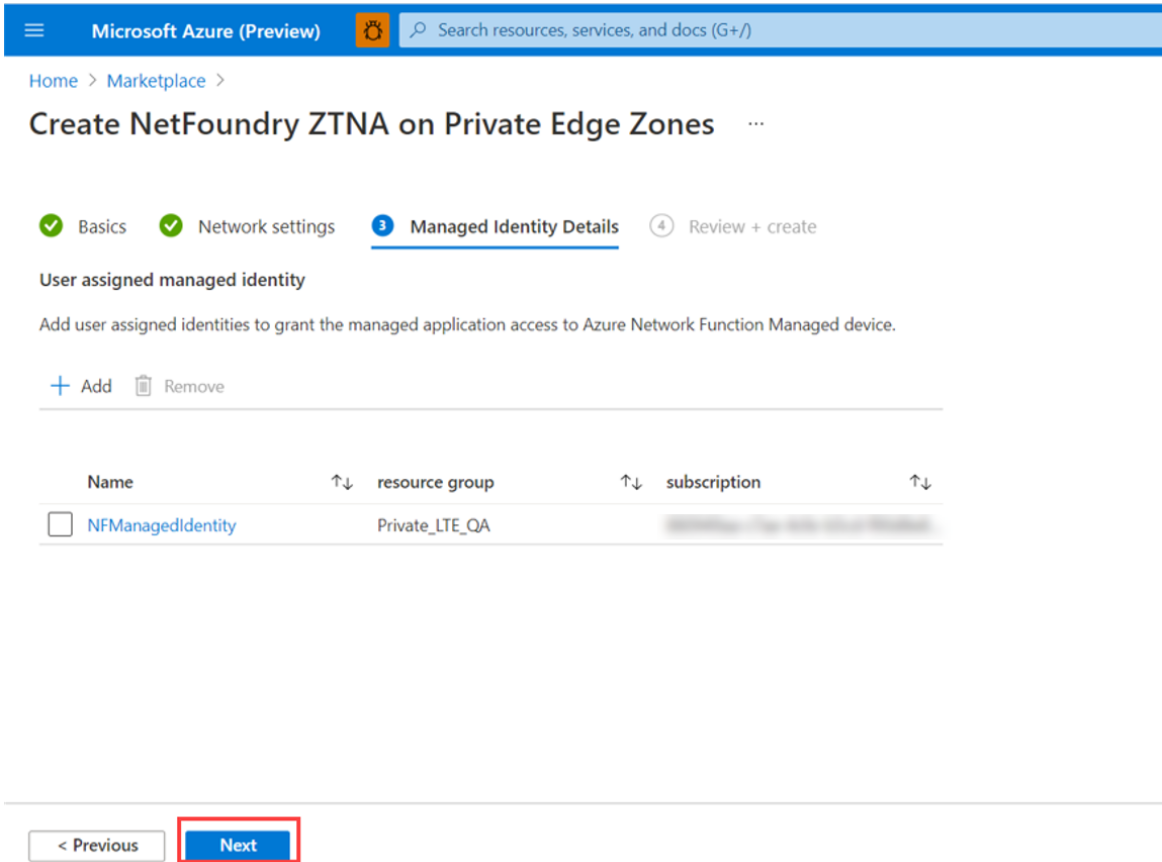
- Resource Group: AzSecPackAutoLontightu
- apns-identity Resource Group: Private_LTE_EMS
- NFManagedIdentity** Resource Group: Private_LTE_QA
- jeffemsha1-identity Resource Group: Private_LTE_EMS

Selected identities:

- NFManagedIdentity** Private_LTE_QA Remove

- Click **Next**.

Figure 3-65 Managed Identity Added



13. Verify the details, select “I agree to the terms and condition above”, and then click the **Create** button.

Figure 3-66 Verifying NetFoundry ZTNA Details

Microsoft Azure (Preview)Report a bug

Home > Marketplace >

Create NetFoundry ZTNA on Private Edge Zones...

Validation Passed

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Co-Admin Access Permission

By checking the box and clicking "Create" I give permission for the template provider referenced above (the "Provider") to have Administrative-level access to one or more Azure resources in order to provide support and management services for the template. In the event of an issue arising from a Provider's services or failure to provide services, your sole recourse is with the Provider. Unless Microsoft is the Provider, Microsoft (I) does not approve, monitor or manage the Provider's access, and (I) bears no responsibility whatsoever for acts or omissions of a Provider.

☐ I agree to the terms and conditions above. *

Basics

Subscription	Private_LTE Dev
Resource group	Private_LTE_ASE
Region	East US
Azure VNF Manager device	ase-pite-6
NetFoundry VNF Name	pite5nf
Edge Router Registration Key	CDG120F2I7
Endpoint JWT	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpVCi99eyJlbSI6Im90dCismV4cCI6MTYyMDk...
Public SSH Key	-
Application Name	pite5netfoundry
Managed Resource Group Name	mrg-application-ziti-private-edge-20210512133949

Network settings

IP address	10.165.32.148
Subnet	10.165.32.0/24
Default gateway	10.165.32.1
DNS 1	-
DNS 2	-

Managed Identity Details

Identity type	UserAssigned
User assigned identity	/subscriptions/880949aa-c7ae-4cfe-b5cd-90d8e81d776/resourceGroups/Pri...

Create

< Previous

Next

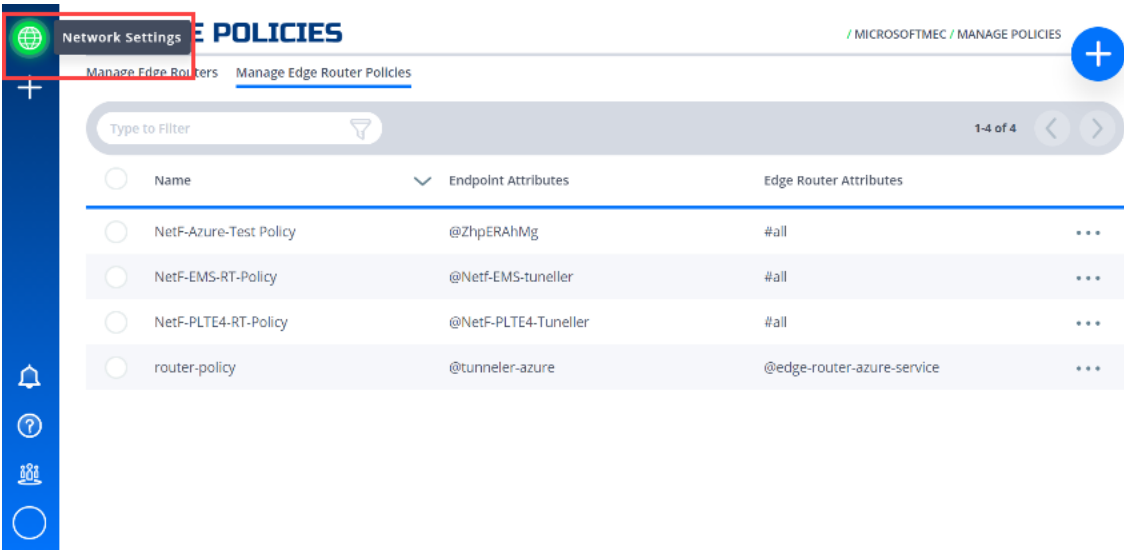
Download a template for automation

Creating Services

To create a service:

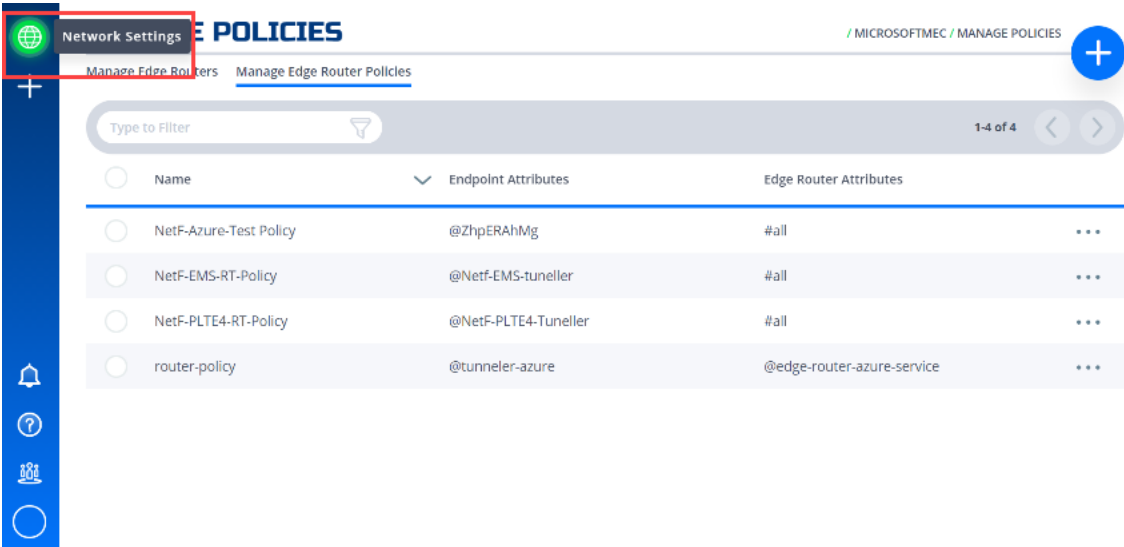
1. Go to top-left corner of the page and click **Network Settings**.

Figure 3-67 Opening Network Settings



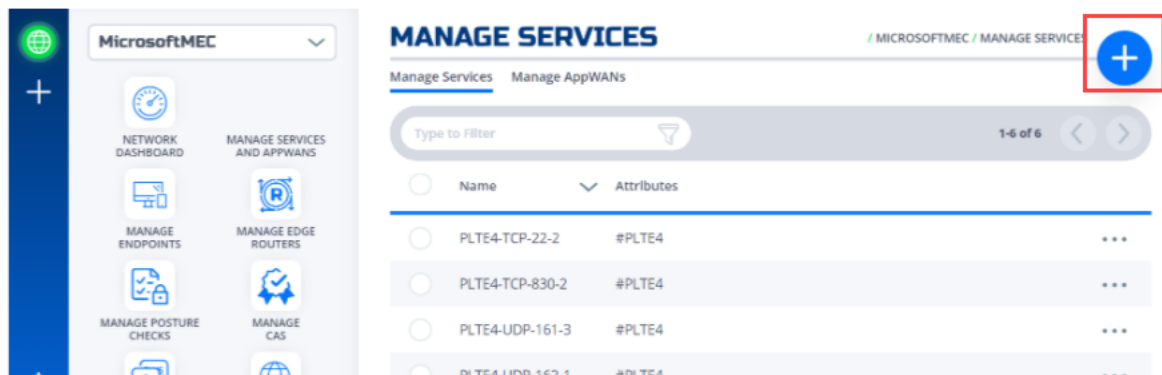
2. Click **Manage Services and AppWANs**.

Figure 3-68 Selecting Manage Services and AppWANs



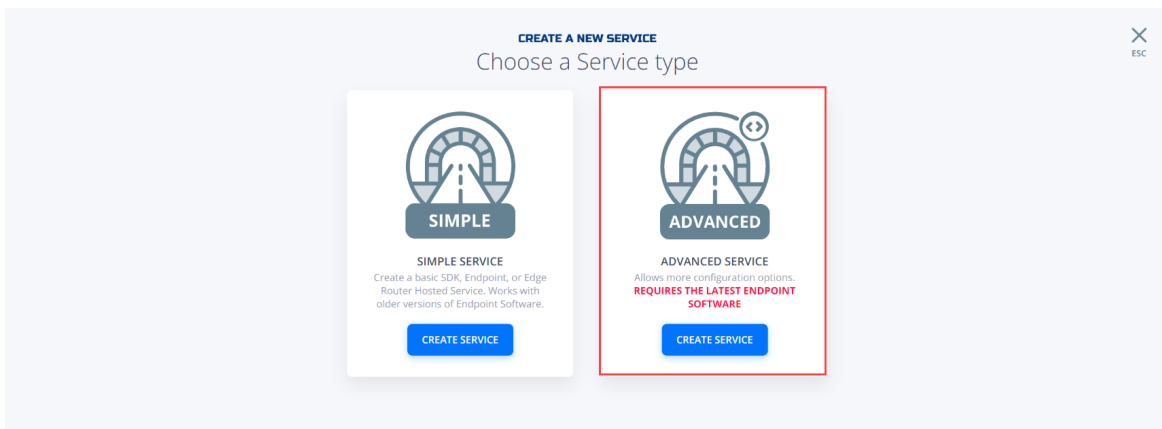
3. Under the **Manage Services** tab, click .

Figure 3-69 Creating Services



4. Select **Advanced Service**.

Figure 3-70 Selecting Advanced Service



5. Provide the following information to create the service for the traffic from ASE to Private Network Service Manager:
 - i. **Service Name:** Provide a name
 - ii. **Service Attributes:** Use #ServiceManager or something similar for all services created for connections towards SM
 - iii. **Edge Router Attributes:** choose the **NetFoundryRouter Attributes** created in the *Prerequisite* section.
 - iv. **END-TO-END ENCRYPTION (Optional):** NO
- **Hosting Strategy**
 - i. **Intercept Host Name/IP:** IP Address of the NetFoundry VM in Azure Stack Edge
 - ii. **Port/Range:** Provide the port/range
 - iii. Keep the status of **Native Application SDK Based** to **No**
 - iv. Select **Endpoint Hosted**.

- v. **Select an Endpoint to Host this Service:** Select the Edge Router created for NetFoundry VM in Azure
- vi. **Protocol:** Select UDP.
- vii. **Host Name / IP:** IP Address of Private Network Service Manager
- viii. **Port:** Type “162”
- ix. Click **Create**.

Figure 3-71 Creating a New Service

CREATE A NEW SERVICE

Enter your service details

[What's a Service?](#)

SERVICE NAME REQUIRED

PLTE10-ASE-CLOUD-162

SERVICE ATTRIBUTES OPTIONAL

Add attributes to group services

✕ #PLTE10_ASE_TO_SM

EDGE ROUTER ATTRIBUTES OPTIONAL

Default: #all, restricts routers available to this service

END-TO-END ENCRYPTION OPTIONAL

ENCRYPT THIS SERVICE ☐ NO

CLIENT CONFIGURATION REQUIRED

INTERCEPT HOST NAME / IP **PORT**

10.165.32.188 162

HOST CONFIGURATION REQUIRED

NATIVE APPLICATION SDK BASED ☐ NO

☒ **ENDPOINT HOSTED** OR ☐ **ROUTER HOSTED**

SELECT ENDPOINTS TO HOST THIS SERVICE

Choose the endpoints to associate to this Service

✕ @Cloud-Edge-Router2

PROTOCOL **HOST NAME / IP** **PORT**

UDP 10.60.0.85 162

CREATE

6. Provide the following information to create the service for the traffic from Private Network Service Manager to ASE:
 - i. **Service Name:** Provide a name.
 - ii. **Service Attributes:** Use #PLTE4 or something similar for all services created for connections towards SM
 - iii. **Edge Router Attributes:** Choose the **NetFoundry Router Attributes** created in the Prerequisites section.
 - iv. **END-TO-END ENCRYPTION (Optional):** NO
- **Hosting Strategy (For the traffic Service Manager to MME/MCC)**

- i. **Intercept Host Name / IP:** IP address of the MCC in Azure Stack Edge. For example, 10.165.2.162
- ii. **Port/Range:** Provide the port/range
- iii. Keep the status of **Native Application SDK Based** to **No**
- iv. Select **Endpoint Hosted**.
- v. **Select a Endpoint to Host this Service:** Select the Edge Router created for NetFoundry VM in Azure Stack Edge
- vi. **Protocol:** Select UDP.
- vii. **Host Name/IP:** IP address of the MCC in Azure Stack Edge. For example, 10.165.2.162.
- viii. **Port:** Type “161”.

Figure 3-72 Editing Service

CREATE A NEW SERVICE

Enter your service details

[What's a Service?](#)

SERVICE NAME REQUIRED

PLTE10-CLOUD-ASE-GW-161

SERVICE ATTRIBUTES OPTIONAL

Add attributes to group services

EDGE ROUTER ATTRIBUTES OPTIONAL

Default: #all, restricts routers available to this service

END-TO-END ENCRYPTION OPTIONAL

ENCRYPT THIS SERVICE ☐ NO

CLIENT CONFIGURATION REQUIRED

INTERCEPT HOST NAME / IP **PORT**

10.165.32.183 161

HOST CONFIGURATION REQUIRED

NATIVE APPLICATION SDK BASED ☐ NO

☒ **ENDPOINT HOSTED** OR ☐ **ROUTER HOSTED**

SELECT ENDPOINTS TO HOST THIS SERVICE

Choose the endpoints to associate to this Service

PROTOCOL **HOST NAME / IP** **PORT**

UDP 10.165.32.183 161

7. Repeat the procedure for the ports given in the following table.

Source	Destination	Port & Protocol	Purpose
Cloud - Service Manager	MME	830 – TCP (SSH)	NETCONF
Cloud - Service Manager	MME	161 – UDP	SNMP
Cloud - Service Manager	MME	22-TCP (SFTP)	File Transfer
Cloud - Service Manager	MME	80-TCP (REST)	RESE API
Cloud - Service Manager	SAEGW	2022 – TCP (SSH)	NETCONF
Cloud - Service Manager	SAEGW	2222 – TCP (SFTP)	File Transfer
Cloud - Service Manager	SAEGW	161 – UDP	SNMP
Cloud - Service Manager	HSS	8883 - TCP (MQTT)	Upgrade, Config Backup, and Audit
Cloud - Service Manager	HSS	1883 - TCP	MQTT
Cloud - Service Manager	HSS	22 - TCP (SFTP)	File Transfer
Cloud - Service Manager	HSS	12272 – TCP	ECPY
NetFoundry instance in ASE	Cloud – Service Manager	162 – UDP	Trap Reception

Trap Reception Configuration on Private Network Service Manager

***Note:** Modify the existing 162 port service with the value provided for the trap reception.*

To receive traps from the VNFs reside at ASE to the Service Manager deployed in the Azure cloud:

1. Enable **Show Transparency** toggle button in Advanced NetFoundry Service.
2. Provide the Management IP Address of NetFoundry ASE in **Allowed Source Addresses** on the VNFs as a trap receiver.
3. Provide the **Forward Address** - Private IP of Service Manager (primary cluster node that has public IP).
4. Click **Update**.

For example, the below given screenshot refers to HSS VNF.

Figure 3-73 Configuring Trap Reception

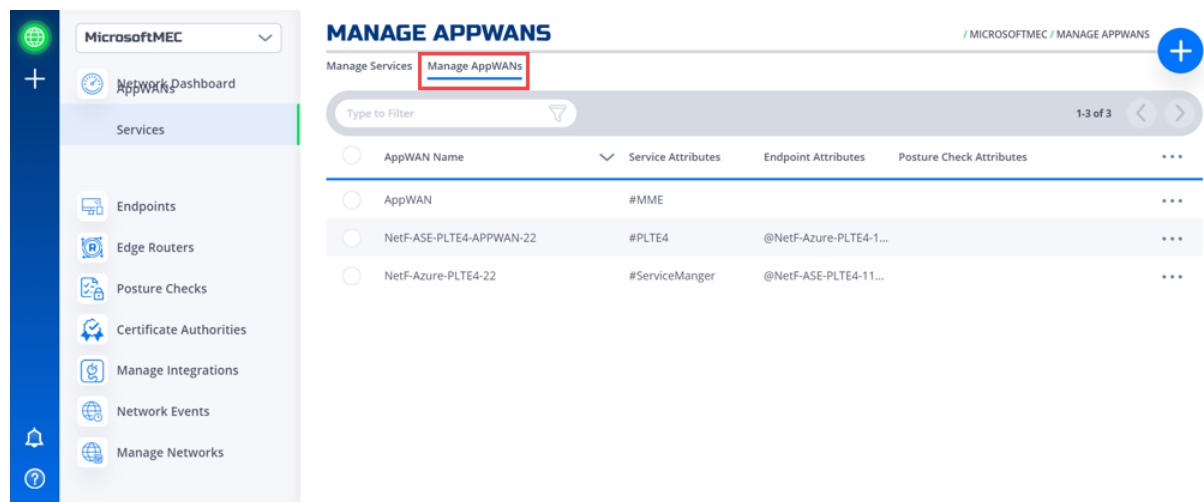
Creating AppWANs

After setting up the services, you can define policies to determine which services the endpoints have permission to access.

To Create AppWANs for the traffic ASE to Private Network Service Manager:

1. Go to top-left corner of the homepage and click **Network Settings** .
2. Go to **Manage Services and AppWANs** and then click the **Manage AppWANs** tab.
3. Create separate App WANs for each ASE box.

Figure 3-74 Selecting Manage AppWANs



4. Go to the top-right corner of the page and click the  button to create a new AppWAN for service attribute created in previous section (**#ServiceManager**) and endpoint of tunneler in ASE.
5. Provide the following AppWAN attributes information:
 - i. **APPWAN NAME:** Provide any name to represent traffic from ASE to Private Network Service Manager
 - ii. **SERVICE ATTRIBUTES:** Choose **#ServiceManager** created in the previous section
 - iii. **ENDPOINT ATTRIBUTES:** Choose the endpoint created for ASE in previous section

Figure 3-75 Entering AppWAN Attributes

6. To create AppWANs for the traffic Private Network Service Manager to ASE, provide the following AppWAN attributes:
 - i. **APPWAN NAME:** Provide any name to represent traffic from Private Network Service Manager to ASE
 - ii. **SERVICE ATTRIBUTES:** Choose **#PLTE4** created in the previous section
 - iii. **ENDPOINT ATTRIBUTES:** Choose the endpoint created for Azure in the previous section

Figure 3-76 Providing AppWAN Details

The screenshot displays the 'EDIT YOUR APPWAN' interface with the title 'Enter your AppWAN details' and a link 'What's an AppWAN?'. The interface includes a left sidebar with four sections: 'APPWAN NAME' (REQUIRED), 'SERVICE ATTRIBUTES' (OPTIONAL), 'ENDPOINT ATTRIBUTES' (OPTIONAL), and 'POSTURE ATTRIBUTES' (OPTIONAL). The 'APPWAN NAME' field contains 'NetF-Azure-PLTE4-22'. The 'SERVICE ATTRIBUTES' section shows a list of services with '#ServiceManger' selected. The 'ENDPOINT ATTRIBUTES' section shows a list of endpoints with '@NetF-ASE-PLTE4-1127-EP' selected. The 'POSTURE ATTRIBUTES' section is empty. On the right, there are three preview panels: 'SERVICES PREVIEW' (7 items), 'ENDPOINTS PREVIEW' (1 item), and 'POSTURE PREVIEW' (0 items). The 'SERVICES PREVIEW' list includes: NetF-Azure-PLTE4-161, NetF-Azure-PLTE4-162, NetF-Azure-PLTE4-2022, NetF-Azure-PLTE4-21, NetF-Azure-PLTE4-2222-1, NetF-Azure-PLTE4-80, and NetF-Azure-PLTE4-830. The 'ENDPOINTS PREVIEW' list includes: NetF-ASE-PLTE4-1127-EP. At the bottom right, there is an 'UPDATE' button.

Upgrading Service Manager

Overview

This section describes the methods to upgrade The Private Network Service Manager applications running on the Azure Cloud platform.

Copying Artifacts from Azure Storage Account

1. To complete az login, execute the following command.

```
az login --identity
```

Note: VM's identity must have read access to the storage container.

2. To get access token for storage account, execute the following command.

```
curl 'http://169.254.169.254/metadata/identity/oauth2/token?api-version=2018-02-01&resource=https%3A%2F%2Fstorage.azure.com%2F' -H Metadata:true | cut -d '"' -f 4
```

3. Use the token to pull the needed files

```
curl -O https://$STORAGE_ACCOUNT.blob.core.windows.net/$STORAGE_CONTAINER/$FILE_NAME -H "x-ms-version: 2017-11-09" -H "Authorization: Bearer $TOKEN"
```

Upgrading the Application

ASAP Standalone Upgrade

No ASAP upgrade is necessary when upgrading to APNS version Release 1.3.0.

ASAP HA Upgrade from ASAP 7.1 to ASAP 7.3

1. Stop the PNSM servers in both the primary and secondary nodes.
2. Login to ASAP primary, login as **root** account or login to **affirmed user** and **sudo su -**

3. Stop the MySQL replication by issuing the following command on both the primary and secondary nodes of the HA cluster.

```
mysql -uroot -p -e "stop slave;"
```
4. Stop the PCS cluster on the primary node.

```
pcs cluster stop-all
```
5. Take the backup of Vaults on the primary node.

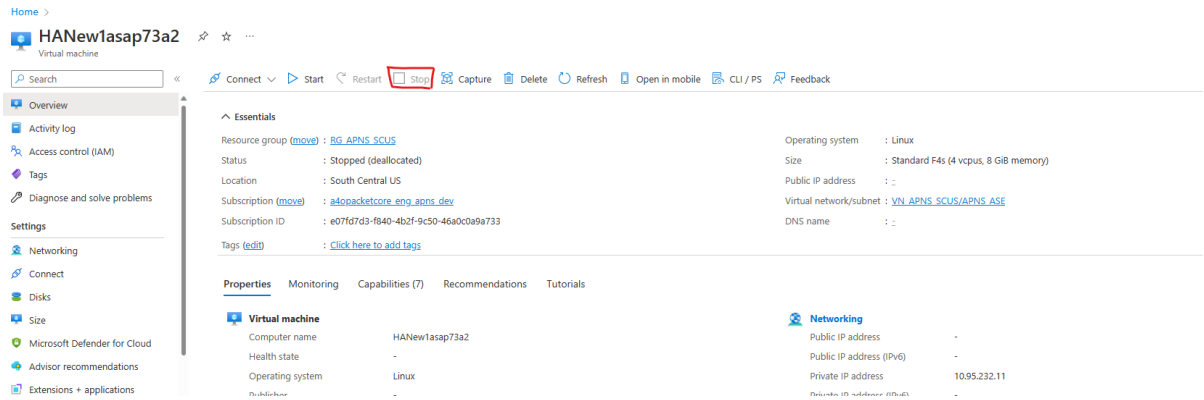
```
cd /opt/affirmed/asap/vault
./vault_export.sh
```
6. Run the following commands to take a database backup. This creates the vTransactor_db_dump.sql and camunda_db_dump.sql files.

```
cd/opt/affirmed/asap/dboperations/bin
sh backupdatabase.sh vTransactor vTransactor_db_dump.sql
sh backupdatabase.sh camunda camunda_db_dump.sql
```
7. Save the vTransactor_db_dump.sql, camunda_db_dump.sql, and the vault_data files externally. Note that you can use one of the PNSM nodes.

```
cd/opt/affirmed/asap/dboperations/bin
scp *sql affirmed@<external_vm_ip>:/home/affirmed/
cd /opt/affirmed/asap/vault
scp ./vault_data affirmed@<external_vm_ip>:/home/affirmed/
```
8. Ensure the files are copied properly and available in the external VM.
9. Delete the old ASAP-related resources from the portal. For example, if the ASAP name was given in the ARM template as **HAOld2asap71** then the list of Azure resources shown below must be removed.

☐ Name ↑↓	Type ↑↓
☐  HAOld2asap71availabilitySet	Availability set
☐  HAOld2asap711_disk1_2312bb49ecca46b4adb7f0b196863aa7	Disk
☐  HAOld2asap712_disk1_841d8135c58f42eda23e641fa924f234	Disk
☐  HAOld2asap71-image	Image
☐  HAOld2asap71loadBalancer	Load balancer
☐  HAOld2asap71-identity	Managed Identity
☐  HAOld2asap711-mgmt-int	Network Interface
☐  HAOld2asap712-mgmt-int	Network Interface
☐  HAOld2asap711	Virtual machine
☐  HAOld2asap712	Virtual machine

10. Once resources are deleted, redeploy ASAP 7.3 using the ASAP HA ARM template, available in the customer FTP site.
11. Ensure the ASAP deployment succeeds and that both ASAP nodes are installed.
12. Stop the secondary ASAP node from the portal if it is running:



13. Login to the primary ASAP node using Bastion or ssh.
14. Copy the backup files to the ASAP primary node.


```
cd home directory
scp affirmed@<external_vm_ip>:/home/affirmed/*.sql ./
scp affirmed@<external_vm_ip>:/home/affirmed/vault_data ./
```
15. Restore the database(s) on the primary node using the following commands.


```
cd /opt/affirmed/asap/dboperations/bin
sudo sh restoredatabase.sh vTransactor
/home/affirmed/vTransactor_db_dump.sql
sudo sh restoredatabase.sh camunda /home/affirmed/camunda_
db_dump.sql
```
16. Use the following command to migrate passwords on the primary node.


```
sudo sh /opt/affirmed/asap/conf/migrate_passwords.sh vault_
data
```
17. On the primary node, start asap-service and verify success in the user interface.
18. Validate the UI in the primary node.
19. On the primary node, stop asap-service.
20. Restart the secondary node (stopped previously in these instructions).
21. On the Secondary node, start and stop the ASAP service once by executing the commands:


```
sudo service start asap-service
sudo service stop asap-service
```
22. Setup password-less SSH between the primary and secondary ASAP nodes.

From primary node : `ssh-copy-id -i .ssh/id_rsa.pub affirmed@<asap_`
`secondary_node_ip>`

From secondary node : `ssh-copy-id -i .ssh/id_rsa.pub affirmed@<asap_`
`primary_node_ip>`
23. Setup HA from the ASAP primary node.


```
sudo /opt/affirmed/asap/cluster/ha_setup.sh <vip> <ipnode1>
<namenode1> <ipnode2> <namenode2> <ipinterface> <database>
```

For example: vip - 10.95.232.12 ; ipnode1 - 10.95.232.10 ; namenode1 - Jan4asap73a1 ;
ipnode2 - 10.95.232.11 ; namenode2 - Jan4asap73a2 ; ipinterface – eth0 ; database -
vTransactor

```
[affirmed@HANew1asap73a1 ~]$ sudo
/opt/affirmed/asap/cluster/ha_setup.sh 10.95.232.12
10.95.232.10 Jan4asap73a1 10.95.232.11 Jan4asap73a2 eth0
vTransactor
```

[sudo] password for affirmed:

Verify the following details:

VIP = 10.95.232.12

ipNode1 = 10.95.232.10

nameNode1 = Jan4asap73a1

ipNode2 = 10.95.232.11

nameNode2 = Jan4asap73a2

ipinterface = eth0

database = vTransactor

Are you sure? [Y/N]: Y

24. Reboot the ASAP nodes once after successful completion.
25. Execute the following command from PNSM server bin directory on the primary node to update ASAP server user's password.

```
./vault_util.sh -p 'affirmed123' -a 'AFFIRMED_ASAP_AFFIRMED_
PASSWORD' -v '<ASAP-PASSWORD-CONFIGURED-ON-ASAP-VM>'
```
26. Start the PNSM servers in both nodes.

PNSM Standalone Upgrade

1. To update the application, use the emsupgrade script:

```
cd /opt/Affirmed/NMS/bin

./emsupgrade /data/affirmed/pnsm-new_pkg_name.rpm
```
2. Tail the log file and look for confirmation the upgrade is complete.

```
tail -f /opt/Affirmed/NMS/log/emsupgrade.log

...

...

2021/03/08 11:43:33 INFO [sub_emsupgrade:main::] EMS Upgrade
is successful.
```
3. To verify that EMS Application is running, execute the following command.

```

cd /opt/Affirmed/NMS/bin

./emsstatus

***** Acuitas Server Status *****

Node : ems-hostname - 10.10.10.10

EMS Package : AffirmedEms-12.0.0.0-10

EMS Process : Running [15745]

MySQL

Database : Running

Server Id : 1

*****

```

HA PNSM Upgrade

Prerequisites

The below mentioned prerequisites must be completed to upgrade from Release 1.1 to 1.2.

- The MySQL **root** user password or the replication user '**acuitas**' password should match the below password constraints:
 - One upper case
 - One lower case
 - One special character
 - One numeric value

If the MySQL **root** user password or the replication user password are not compliant with the above mentioned password criteria in the PNSM 1.1 release, it must be reset to new password to meet the password criteria before upgrading to 1.2.

To reset the password of root user and replication user:

1. Stop the server if it is running in both servers.


```
cd /opt/Affirmed/NMS/bin/
./emsha stopAll
```
2. Run the following command to remove the mylogin.cnf file in both servers


```
rm -rf /home/affirmed/.mylogin.cnf
```
3. Run the below script to change the root user and replication user's password of Database with a new password as per the password criteria.


```
cd /opt/Affirmed/NMS/bin
./configureserver.sh
```

Note: Provide the new respective passwords when it is prompted for root and replication user.

4. Run the following commands in both primary and secondary nodes.


```
mysql_config_editor set --login-path=client --user=root --
host=localhost --password

mysql_config_editor set --login-path=replication --
user=acuitas --password
```
5. Run the below command as a root user (**su - root**) in both nodes.


```
mysql_config_editor set --login-path=client --user=root --
host=localhost --password
```
6. Re-configure the HA nodes using the below command in the primary node where the public IP is bounded.

configure_ha_az **--virtualhostname** (name of the active server) **--virtualip** (public ip of active node) **--primaryhostname** (name of active server) **--primaryip** (private ip of active node) **--secondaryhostname** (name of standby node) **--secondaryip** (private ip of standby node) **--ipinterface** **eth0**(use ip a cmd to find the interface) **--netmask** (subnet) **--force** **--resourceGroup** (resourcegroupname)

Sample Command:

```
./configure_ha_az --virtualhostname hapnsm --virtualip
137.116.91.137 --primaryhostname hapnsm2 --primaryip
10.60.0.25 --secondaryhostname hapnsm1 --secondaryip
10.60.0.24 --ipinterface eth0 --netmask 24 --force --
resourceGroup LTE_RESOURCE_GRP_NAME
```

7. Run the below script to check the server status on both nodes once the HA cluster is configured successfully.


```
cd /opt/Affirmed/NMS/bin

./emsstatus
```

Upgrade Process

If the upgrade does not require a major upgrade to the OS, you can use the standard upgrade procedure. Use the haupgrade script to upgrade PNSM on an HA pair.

The haupgrade script runs in the background. The script prompts for input and then displays the message:

The command \$COMMAND is running in the background.

Check log file \$LOG_FILE for the command result.

When you use the **--client** option, the process runs in the foreground. Interrupting the script with **<CTRL>-C** terminates the operation, and is not recommended. The process will not continue to run in the background.

To upgrade the software with a newer version:

1. On both servers, copy the upgrade package `pnsmlha-new_pkg_name` to `/data/affirmed` (using `wget` “SASURL of the pkg”), where `pnsmlha-new_pkg_name` is the new Acuitas package.
2. Log into either of the HA servers as user account **affirmed** using the physical IP address (*not* the virtual IP address).
3. On either of the HA servers, issue the following commands:


```
cd /opt/Affirmed/NMS/bin

./haupgrade /data/affirmed/pnsmlha-new_pkg_name.rpm

Do you want to upgrade the HA pair on primary <server name>
and secondary <server name> using /data/affirmed/ pnsmlha-
new_pkg_name.rpm (Y/N)?
```
4. Enter **y** to continue with the upgrade.


```
Upgrade will start by upgrading the standby node <server
name>. Continue (Y/N)?
```
5. Enter **y** to continue with the upgrade. The upgrade can take some time to complete. When the upgrade completes, you can check the log file for the status.

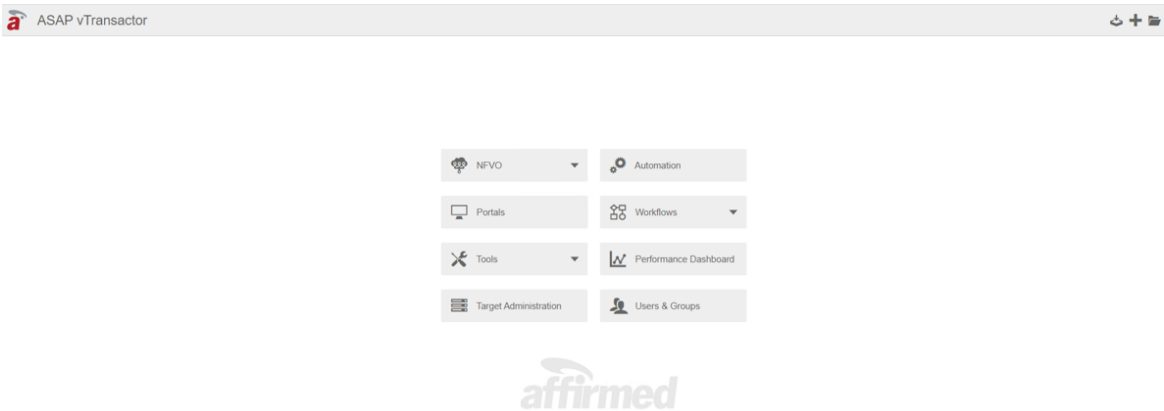
***Note:** Once the Service Manager is successfully upgraded, do not perform any operations (such as SIM operations, NE backup, etc) on the application. Any operation should only be performed after all the deployed networks are upgraded.*

Post-Upgrade Validation**Validating Service Model Upgrade**

Perform the following validation steps:

1. Log in to the ASAP Web GUI.

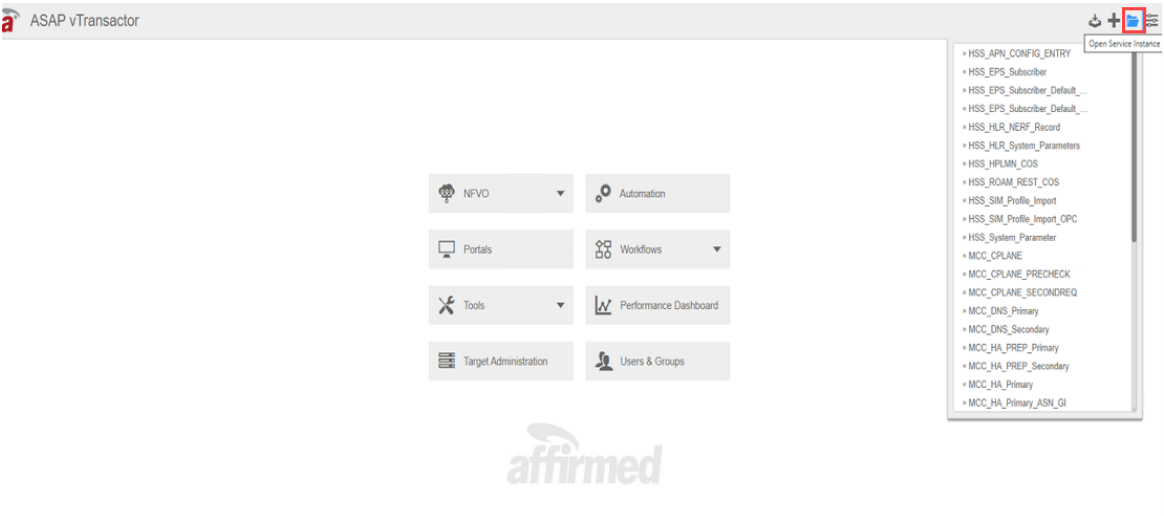
Figure 4-1 ASAP Web GUI



2. Go to the top-right corner of the page, click **Open Service Instance** to get the list of service models imported in the ASAP Server and check whether the PNSM Service Models available in `/opt/Affirmed/NMS/server/ems/resources/service_models/` are imported to the ASAP server.

Note: To go to PNSM service models path, log into PNSM CLI:
`ssh <PNSM_CLI_USERNAME>@<PNSM_IP_ADDRESS>`

Figure 4-2 Opening Service Instance



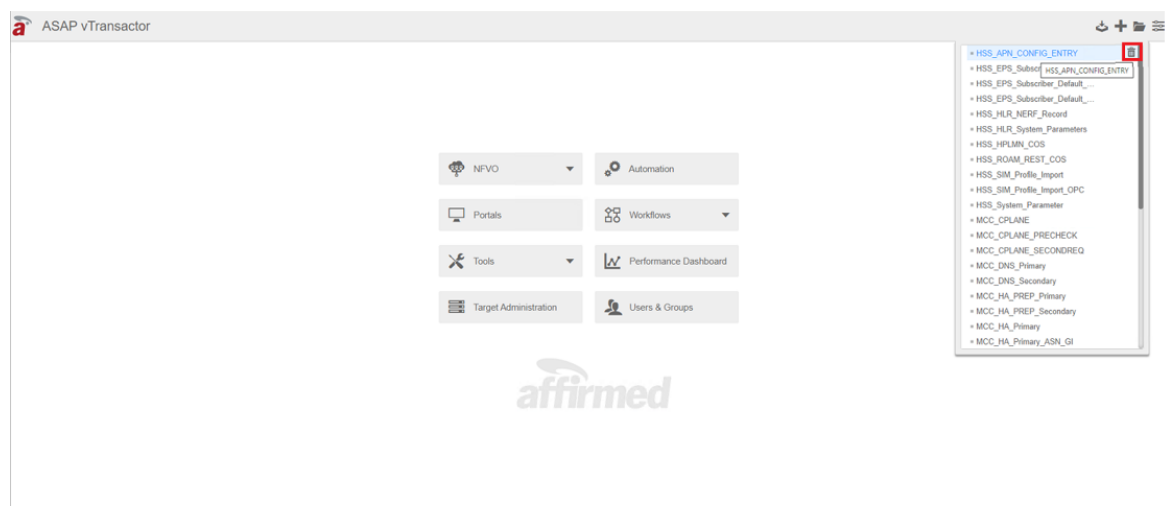
Note: If any service models are not imported to ASAP Server, delete all the existing service models from the ASAP Server and import all the PNSM 1.2 Service Models in the ASAP Server

Deleting Existing Service Model Upgrade

To delete:

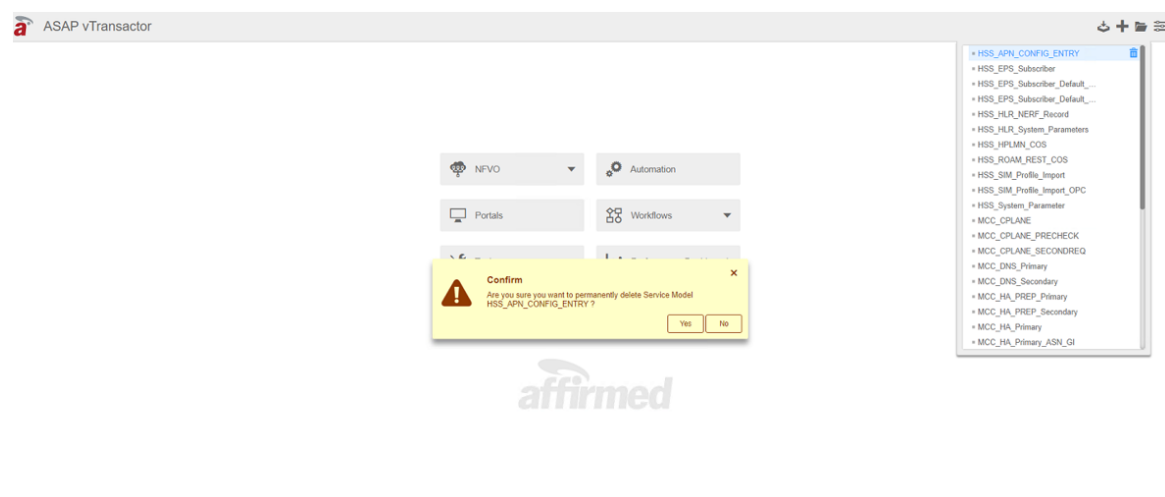
1. Click **Open Service Instance** icon in the top right corner to get the list of services models imported in the ASAP server and hover mouse on each service model in the list to get the **Delete** option against the corresponding Service Model. Click **Delete**

Figure 4-3 Deleting Service Model

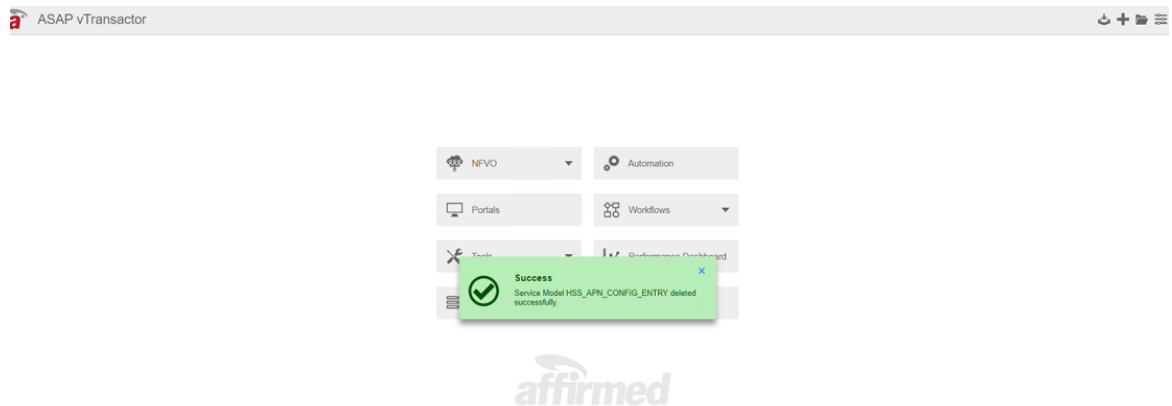


2. You will receive ASAP Server confirmation message. Click **Yes** to proceed.

Figure 4-4 ASAP Server Confirmation Message



3. After successful deletion, success message with the service models name will be prompted.

Figure 4-5 Service Model Deletion

Note: Make sure all the existing service models are deleted from the ASAP Server before importing the service models.

Importing the Service Models to the ASAP Server

To import:

1. Log into the PNSM CLI.

```
ssh <PNSM_CLI_USERNAME>@<PNSM_IP_ADDRESS>
```
2. Execute the following command to copy the Service Models from the PNSM Server (/opt/Affirmed/NMS/server/ems/resources/service_models/) to the ASAP Server (ASAP server path - /home/affirmed)

```
scp /opt/Affirmed/NMS/server/ems/resources/service_models/*  
<ASAP_SERVER_CLI_USERNAME>@<ASAP_SERVER_IP_ADDRESS>:/home/affirmed/
```
3. To continue, provide ASAP_SERVER_CLI_PASSWORD.
4. On successful copy of all the Service Models, execute the CURL IMPORT commands to import the PNSM 1.2 Services Model to the ASAP Server.

CURL command to import the service models

```
curl -u <ASAP_SERVER_WEB_GUI_USERNAME>:<ASAP_SERVER_WEB_GUI_PASSWORD> -X GET http:// <ASAP_SERVER_IP_ADDRESS>:8080  
/asap/rest/nb/serviceModel/import?fileName=%2Fhome%2FAffirmed%2F<  
SERVICE_MODEL_ZIP_FILE>
```

Sample IMPORT Request


```
curl -u <ASAP_SERVER_WEB_GUI_USERNAME>:<ASAP_SERVER_WEB_GUI_PASSWORD> -X GET http:// <ASAP_SERVER_IP_ADDRESS>:8080/asap/rest/nb/serviceModel/import?fileName=%2Fhome%2Faffirmed%2FHSS_APN_CONFIG_ENTRY.zip
```

Sample IMPORT Response

```
{ "header":
{ "siteId":null,"locale":null,"version":"1","type": "
ASAP", "timestamp":1637679405867}, "body":
{ "result":true, "userDefined": [], "serviceName": "HSS_APN_CONFIG_ENTRY", "version": "1" }}
```

Note: Make sure all the Service Models are imported to the ASAP Server.

Note: CURL GET command must return all the mentioned Service Models on successful import.

Figure 4-6 Sample GET Request

```
curl -u <ASAP_SERVER_WEB_GUI_USERNAME>:<ASAP_SERVER_WEB_GUI_PASSWORD> -X GET http:// <ASAP_SERVER_IP_ADDRESS>:8080/asap/rest/nb/serviceModel/list/getAllServiceModels
```

Sample GET Response

```
{ "header":
{ "siteId":null,"locale":null,"version":"1","type": "
ASAP", "timestamp":null}, "body":
{ "result":true, "serviceModels": [{"serviceName": "HSS_System_Parameter"}, {"serviceName": "HSS_EPS_Subscriber_Default_Record"}] }
```

Note: The sample mention above is just for reference purpose. The GET Response will vary depending upon the number of service models.

Installing Security Patches

Prerequisites

Applications must be taken offline before applying the Security Patch and reboot the OS. The patch is applied to each server independently. If your system is an HA Cluster currently providing service:

1. Stop the Standby server.
2. Patch it.

3. Reboot it manually.
4. Switch over the active role to the freshly patched instance.
5. Verify that normal operation resumes before patching the remaining server.

ASAP

1. Switch to the root user
sudo su - # from the admin user
su - # from the affirmed user
2. Stop the application process.
service asap-service stop
3. Unpack the security patch file and run the installer.
tar xvf an-ems-3.1.0-patch-1.x.y.tgz
nohup ./repo/install.rb > patch.log &
4. Tail the log file and verify the installer completes successfully.
tail -f patch.log

...

...

patch-1.x.y has been installed.
If a newer kernel has been installed please reboot the server!

OS Image Patch

1. Copy the patch file to the Acuitas server in a location with enough space, as the patch files are ~2GB once expanded. For example, if using the /data/affirmed/ directory:
scp ~/Downloads/an-ems-3.1.0-patch-1.4.0.tar.gz
affirmed@<Acuitas_server_IP>:/data/affirmed
2. As affirmed user, stop the EMS process:
cd /opt/affirmed/NMS/bin

For HA systems:
./emsha stopall

For stand-alone systems:
./emsstop
3. Expand the patch file (run as root user):
su -
cd /data/affirmed
tar xvf an-ems-3.1.0-patch-1.4.0.tar.gz

A local repository is created in /data/affirmed:
an-ems-3.1.0-patch-1.4.0.tar.gz repo
4. Run the patch installer script as the root user:

```
cd /data/affirmed && nohup ./repo/install.rb > patch.log &
```

The following output appears, identifying the background job:

```
[1] 28593
```

```
nohup: ignoring input and redirecting stderr to stdout
```

The patch installation process takes approximately 10 minutes and runs in the background.

When the installation completes successfully, you see the Done notification:

```
[1]+ Done nohup ./repo/install.rb > /data/affirmed/patch.log
```

If there is a problem running the job, you see the Exit notification instead:

```
[1]+ Exit 1 nohup ./repo/install.rb > patch_install-1.log
```

You can monitor the installation process at any time by tailing the patch.log file (notifications may be suppressed):

```
tail -fpatch.log
```

5. Because the patch includes a kernel update, reboot the server:

```
reboot
```

6. If upgrading from an OS Patch earlier than OS Patch 1.1.0, you must update your password, as new strength requirements now exist.

Passwords must:

- Contain at least one of each of the characters: upper case letter, lower case letter, numeric, and special character (such as !, @, #, \$, %, ^ & *, ?, =,+,-)
- Be at least 14 characters in length
- Not repeat any of the previous 12 passwords
- Be sufficiently different (at least 5 characters different) from the previous password
- Not be dictionary words or usernames

Once the reboot completes, the APNS server will be in service and running the new patch.

Deploying APNS in Cloud

Overview

There are two ways to deploy APNS, namely:

- Using All-in-Cloud deployment
- Using Distributed Edge deployment (or CUPS deployment).

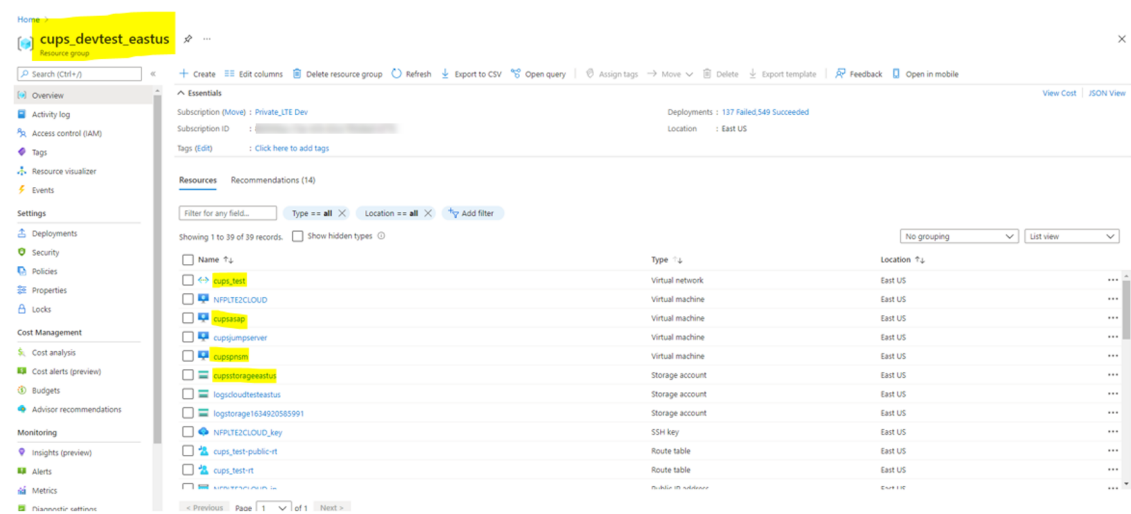
You can choose only one of the above mentioned deployment process.

Prerequisites

In addition to basic parameters, the below given parameters are also required:

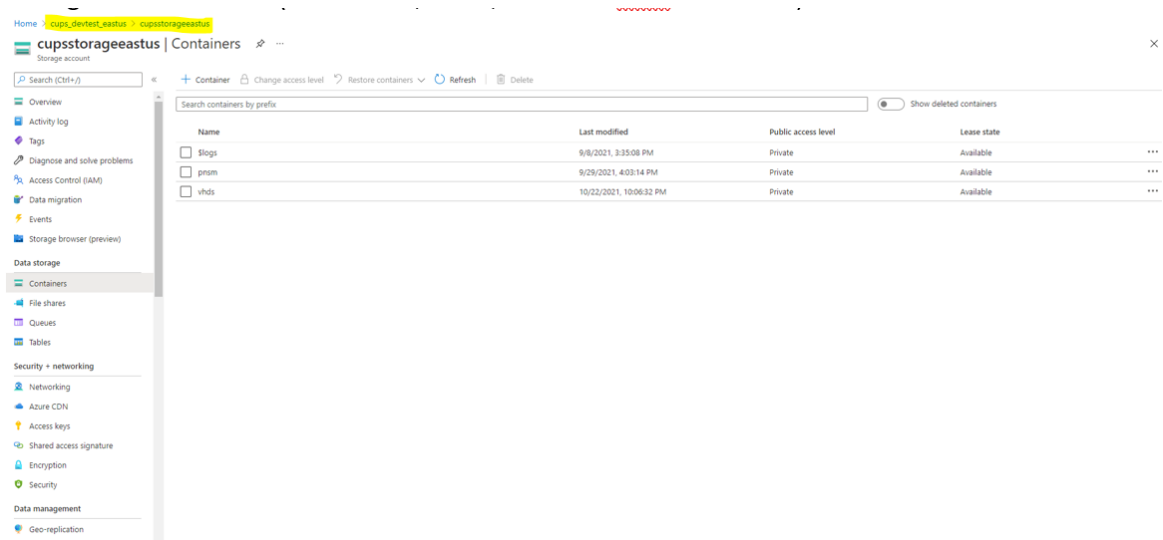
1. Resource Group Name (should be same as Service Manager Resource Group)

Figure 5-1 Sample Resource Group Name



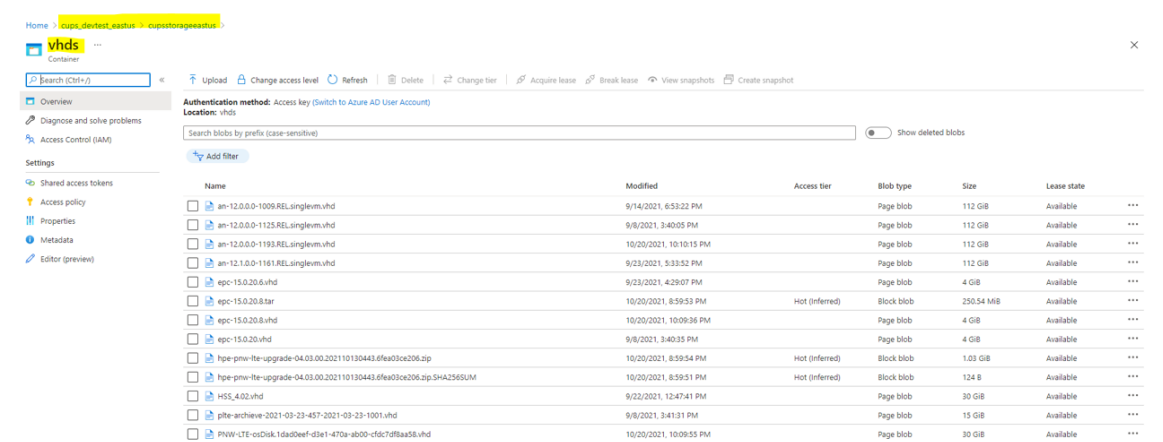
2. Storage Account Name (where HSS, MME, and MCC vhd's are stored)

Figure 5-2 Sample Storage Account Name



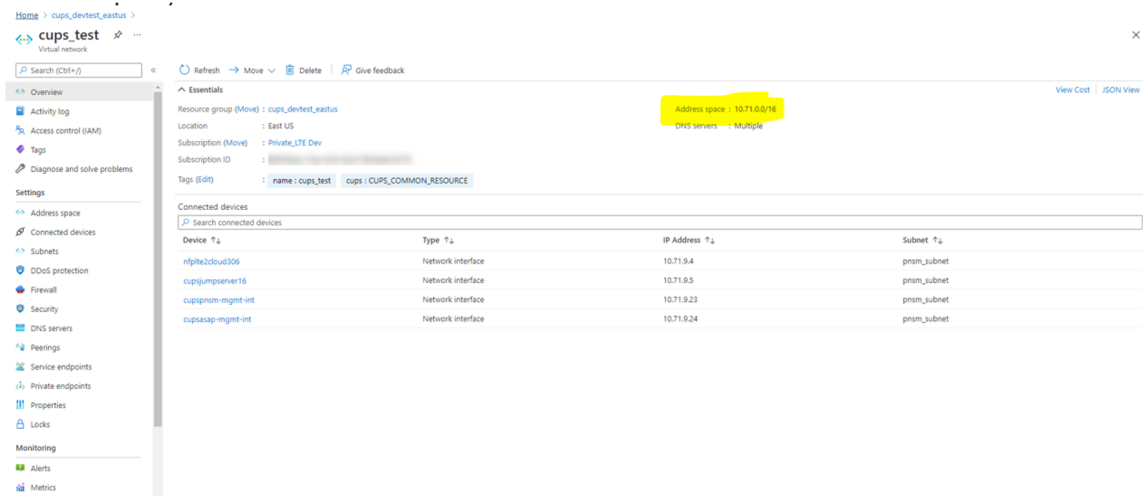
3. Storage Container Name (container name is present in the Storage Account Name)

Figure 5-3 Sample Storage Container Name



4. Virtual Network (VNet) Name (should be under Service Manager resource group with the same address space)

Figure 5-4 Virtual Network Name



PNSM and ASAP servers should be installed with above mentioned virtual network – **Address space IP Address** - 10.71.0.0/16; otherwise, resource communication will not work.

All the IP Addresses for MME/HSS/SAEGW will be system generated. Service Manager will trigger the deployment for HSS, MME, and SAEGW in parallel. So, if the deployment for HSS has failed, still the service manager will deploy the MME and SAEGW.

RSA keys will be generated internally and used for Control Plane deployment. You can download RSA key from Location Management with location name *CUPS_DEFAULT_LOCATION*.

You can create VNet manually. The other remaining resources can be created via Service Manager.

The below mentioned resources will be deployed using the ARM template based deployment on cloud.

- HSS
- SAEGW
- MME

Deploying Fully Cloud

To deploy the Fully Cloud network, refer to Fully cloud section in *Private Network Service Manager User's Guide*.

Deploying Distributed Edge

In addition to above mentioned prerequisites, perform these additional prerequisites:

CUPS NetFoundry Setup

VNet Creation

If the CUPS feature needs to be validated, the VNET should be deployed with below mentioned CIDR. User will not have control to modify the IPAddresses in CUPS deployment, so the following IP layout needs to be used.

VNET and pnsn-subnet can be created by PNSM ARM template or manually. Rest of the subnets are created automatically by the CUPS deployment. Auto subnets are created in **Create Network** and deleted in **Delete Network**

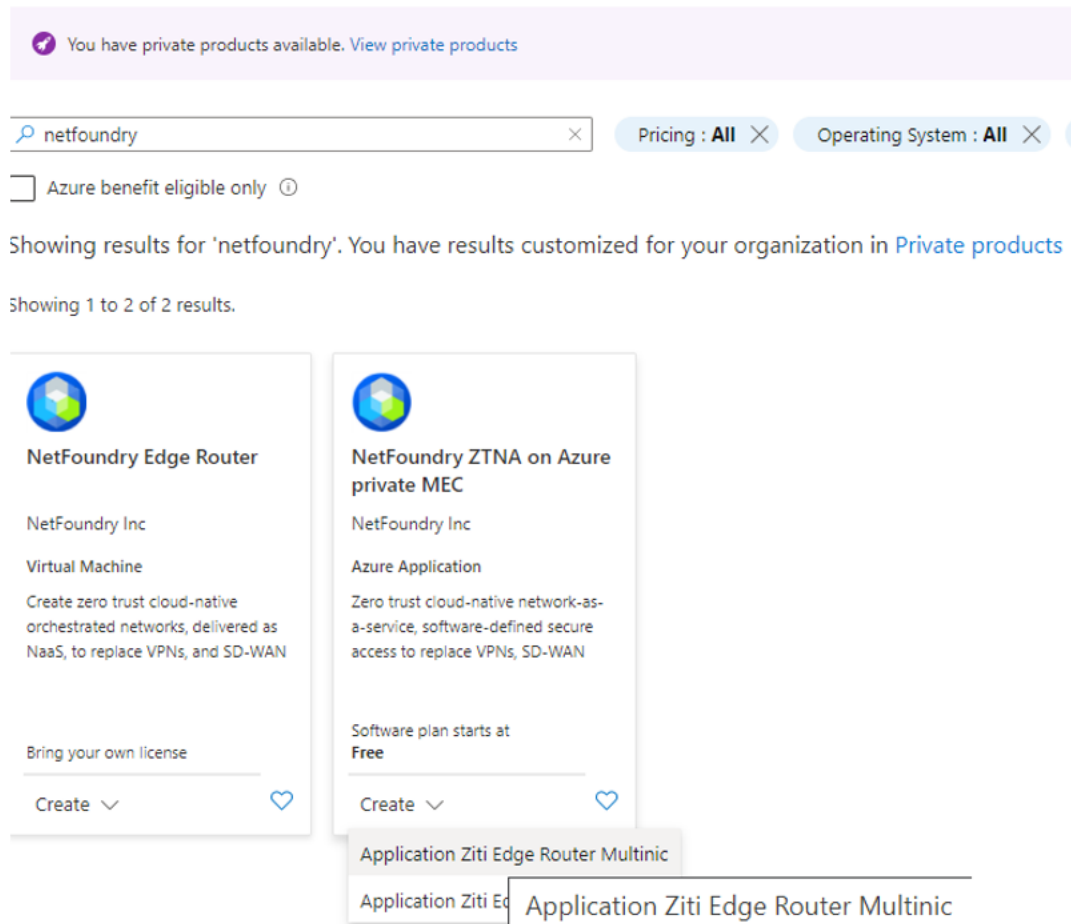
Table 5-1 CUPS Subnet Layout

Subnet	IP Address	Installation Type
vnet	10.71.0.0/16	Manual/PNSM Installation
Pnsn-subnet	10.71.9.0/24	Manual/PNSM Installation
smvhdseastus2-vnet-mgmt	10.71.1.0/24	Auto
smvhdseastus2-vnet-ns1	10.71.4.0/24	Auto
smvhdseastus2-vnet-ns2	10.71.5.0/24	Auto
Plte-data	10.71.3.0/24	Auto
Plte-sig	10.71.7.0/24	Auto
Plte-base	10.71.2.0/24	Auto
Public	10.71.6.0/24	Auto

NetFoundry VM Deployment in Azure Stack Edge for CUPS

To install ASE:

1. Go to Azure portal, type "Marketplace" in the search bar and press **Enter**.
2. Type "netfoundry" in the search bar, go to NetFoundry ZTNA on Azure private MEC, click **Create**, and then select **Application Ziti Edge Router Multinic**.

Figure 5-5 Selecting Ziti Edge Router Multinic

- Under the **Network Settings** tab, provide the require information.

[Home](#) > [Marketplace](#) >

Create NetFoundry ZTNA on Azure private MEC ...

✓ Basics **2 Network settings** ③ Managed Identity Details ④ Review + create

Management

IP address * ⓘ	10.165.32.168	✓
Subnet * ⓘ	10.165.32.0/22	✓
Default gateway * ⓘ	10.165.32.1	✓
DNS 1 ⓘ	10.168.2.193	✓
DNS 2 ⓘ		

Lan

IP address * ⓘ	10.165.61.26	✓
Subnet * ⓘ	10.165.60.0/22	✓
Default gateway ⓘ	10.165.60.1	✓

4. Create the NetFoundry VM. Provide the following values:
 - **Allow Source Address:** 10.71.4.63 and 10.71.4.62 (These are Azure Cloud IP and it will not change)
 - **Intercept IP/Intercept Ranges:** 10.165.61.194 (It will change based on the ASE device)

Figure 5-6 Creating NetFoundry VM

Edit your service details
What's a Service? ⓘ

SERVICE NAME REQUIRED

PLTE8-CLOUD-ASE-CUPS-8805

SERVICE ATTRIBUTES ⓘ OPTIONAL

Add attributes to group services

✕ #PLTE8_SM_TO_ASE

EDGE ROUTER ATTRIBUTES ⓘ OPTIONAL

Default: #all, restricts routers available to this service

END-TO-END ENCRYPTION ⓘ OPTIONAL

ENCRYPT THIS SERVICE ☐ NO

SOURCE TRANSPARENCY ⓘ OPTIONAL

ENABLE SOURCE TRANSPARENCY ☒ YES

ALLOWED SOURCE ADDRESSES

10.71.4.63 ✕ 10.71.4.62 ✕

CLIENT INTERCEPT CONFIGURATION ⓘ REQUIRED

PROTOCOL(S)

☐ TCP ☒ UDP

INTERCEPT IP / INTERCEPT RANGES

10.165.61.194 ✕

PORT / RANGES

8805 ✕

CLIENT CONFIG PREVIEW

Allow Ingress traffic to ports 8805 from 10.165.61.194 via udp

DESTINATION/APP CONFIGURATION ⓘ REQUIRED

SELECT ENDPOINT(S) ⓘ

Choose endpoints or leave blank for #all.

✕ @NF-ASE-PLTE8-MULTINIC2

FORWARD PROTOCOL ☒ YES

☒ UDP

FORWARD ADDRESS ☒ YES

10.165.61.194

FORWARD PORT ☒ YES

8805

If you are deploying using Distributed Edge deployment, refer to Distributed Edge section in *Private Network Service Manager User's Guide*.

Manual Routing

To add the manual route to ASE-NF VM, execute the following command:

```
sudo ip route add 10.165.61.194/32 via 10.165.61.19
```

where, 10.165.61.194 is Uplane PGW_Loop address

10.165.61.19 is Uplane GN Interface IP

To add manual routing to MCC, execute the following command:

```
ip-route 10.71.4.0 24 10.165.61.26
```

where, 10.71.4.0 is CPLANE Subnet

10.165.61.26 is NF ASE LAN IP Address

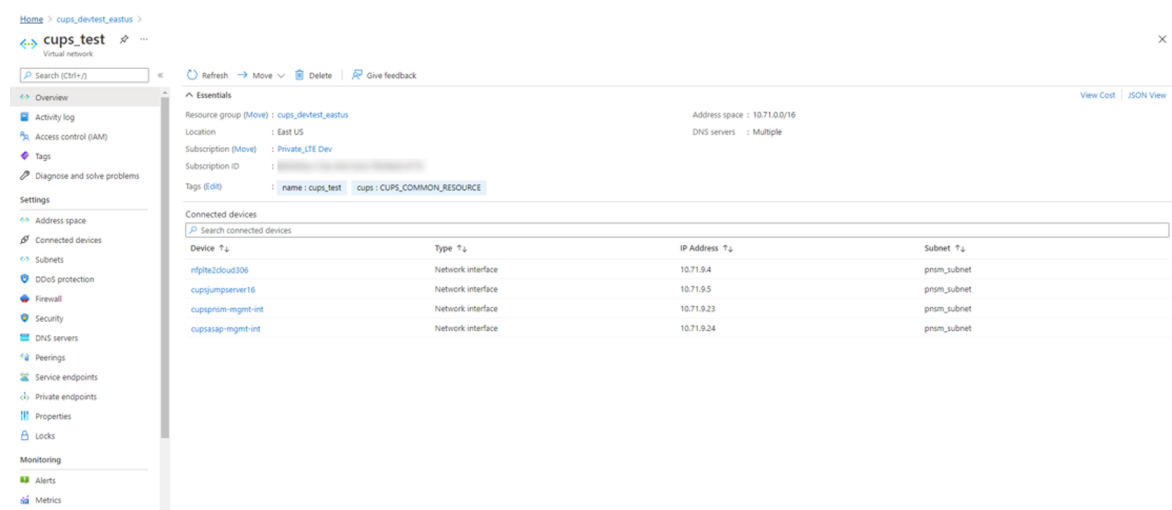
Note: The LAN IP Address for the NF needs to be planned across the lab to avoid overlaps. Need to use the unused IP's from 10.165.61.0/27.

Prerequisite for Cloud Deployments

In addition to previously mentioned prerequisites, ensure the below mentioned prerequisites are also completed.

- Creating Virtual Network with address space - 10.71.0.0/16

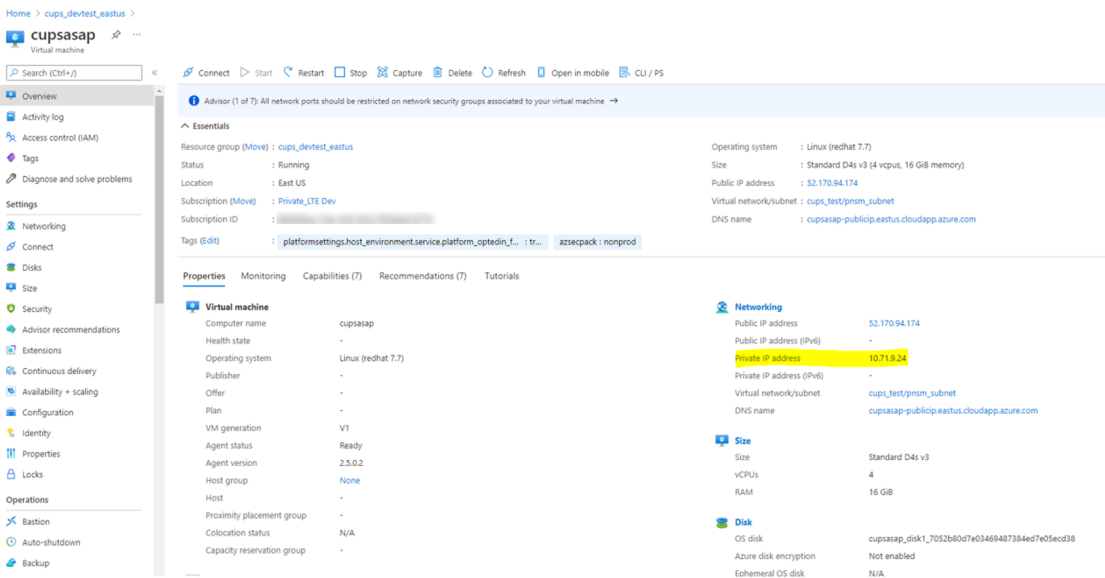
Figure 5-7 Creating Virtual Network



- Install PNSM and ASAP server in the above created resource group with IPs in the above mentioned Virtual Network Address Space.

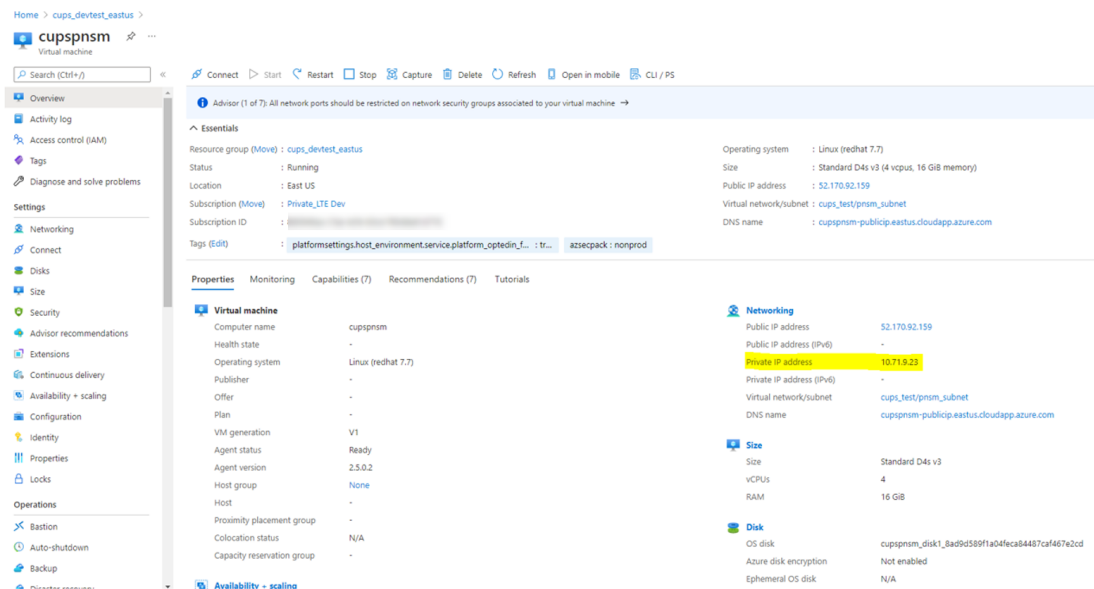
ASAP Server:

Figure 5-8 Sample ASAP Server



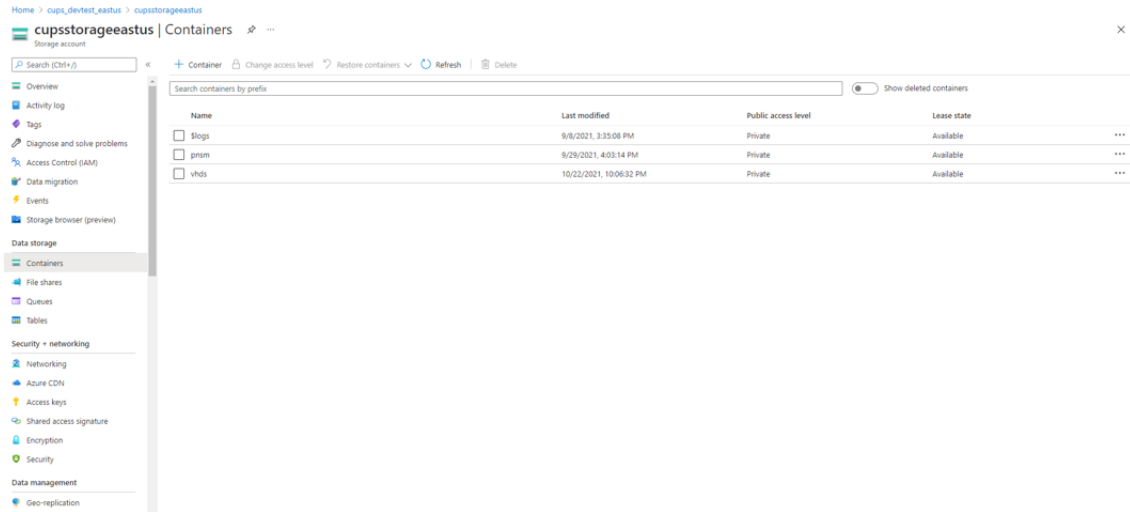
PNSM Server

Figure 5-9 Sample PNSM Server



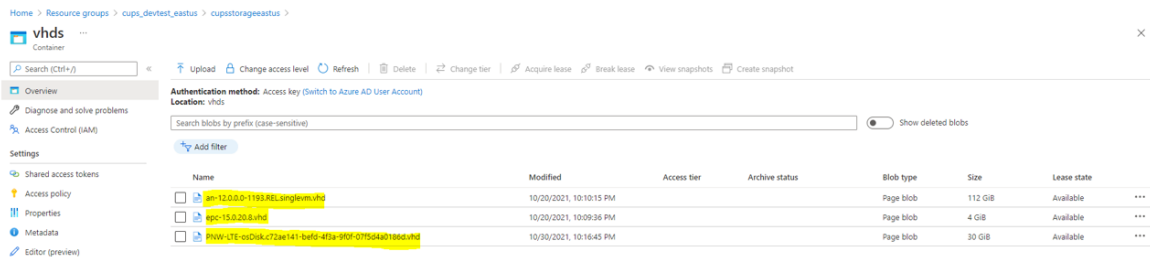
- **Creating Storage Account:** Create a storage account in above create resource group to store the vhd images used for MCC, MME, and HSS deployments. These vhds are available as part of APNS 1.1 release artifacts available in the SFTP server folder.

Figure 5-10 Creating CUPS Storage Account



- **Creating Storage Container:** Create storage container inside above created storage account and copy all necessary vhds inside this container.

Figure 5-11 Creating Storage Container



Azure Cloud: NetFoundry

To install the NetFoundry in the pnsn-subnet:

1. Add route for ASE MGMT with next hop as NF VM.
2. Add route for CPLANE to UPLANE with next hop as NF VM. Any packet going to **10.165.61.194** (uplane) has next hop as NF VM.

Adding routes

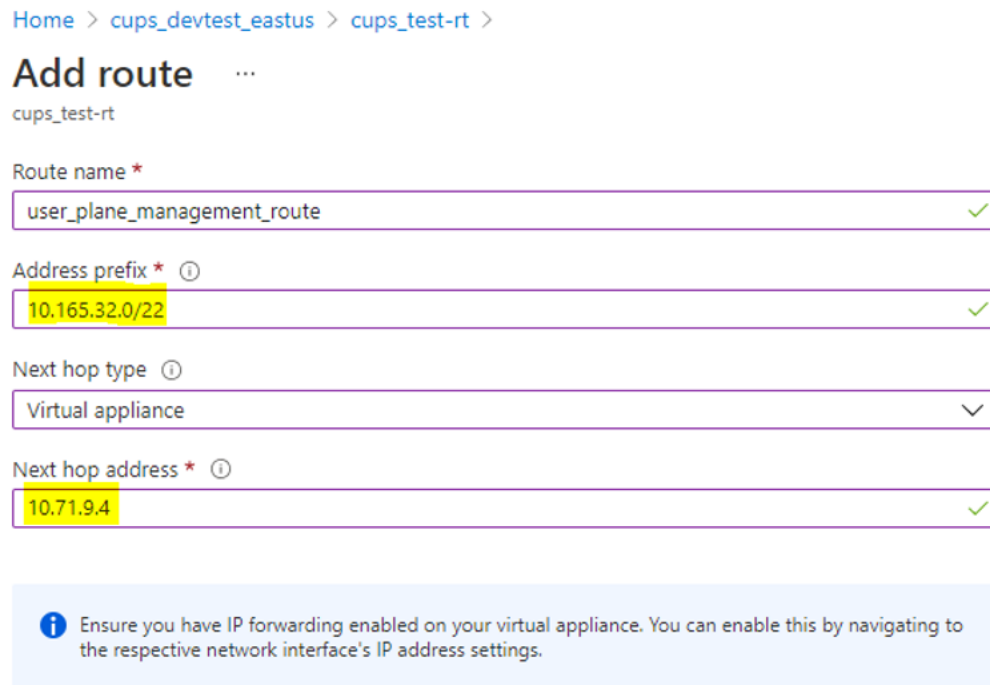
Management Route

Management route needs to be added in Azure portal after network deployment in PNSM. This route is mandatory for adding user plane MCC in PNSM server. If this route is missed, user plane MCC communication will not work properly.

Add a route in Azure portal during Network Creation process. A route table will be created as `<virtual-network-name>-rt` under given resource group.

In the below given figure, `cups_devtest_eastus` is the user given cloud resource group and `cups_test` is the user given virtual network, and `cups_test-rt` is created by Create PNSM operation.

Figure 5-12 Adding Route



Home > cups_devtest_eastus > cups_test-rt >

Add route ...

cups_test-rt

Route name *

user_plane_management_route ✓

Address prefix * ⓘ

10.165.32.0/22 ✓

Next hop type ⓘ

Virtual appliance ✓

Next hop address * ⓘ

10.71.9.4 ✓

i Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.

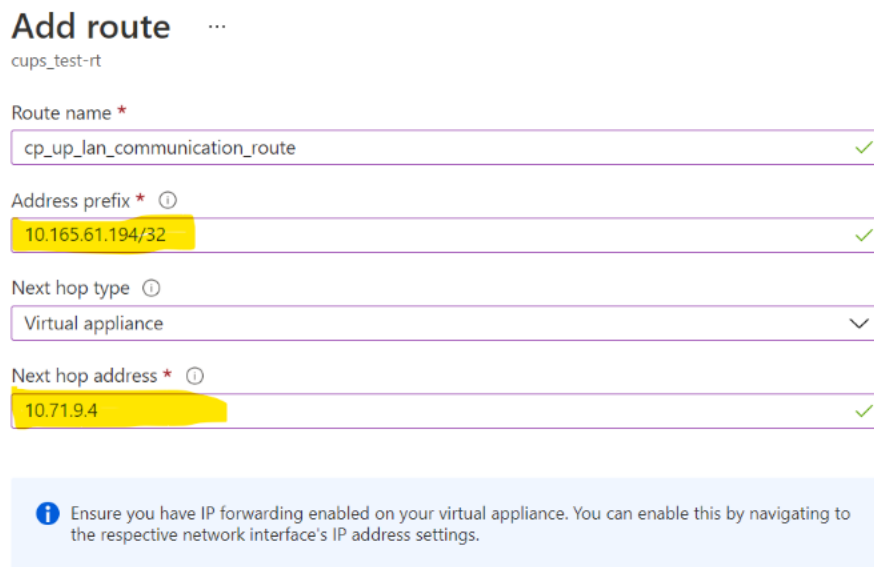
10.165.32.0/22 – Management subnet of User Plane MCC

10.71.9.4 – Cloud Netfoundry Virtual Machine Private IP

LAN Route

LAN route needs to be added after service configuration push from PNSM. This will make the MCC control plane and user plane communication up.

Figure 5-13 Adding LAN Route



Add route ...

cups_test-rt

Route name *

cp_up_lan_communication_route ✓

Address prefix * ⓘ

10.165.61.194/32 ✓

Next hop type ⓘ

Virtual appliance ✓

Next hop address * ⓘ

10.71.9.4 ✓

i Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.

10.165.61.194/32 – PGW address of User Plane MCC

10.71.9.4 – Cloud NetFoundry Virtual Machine Private IP